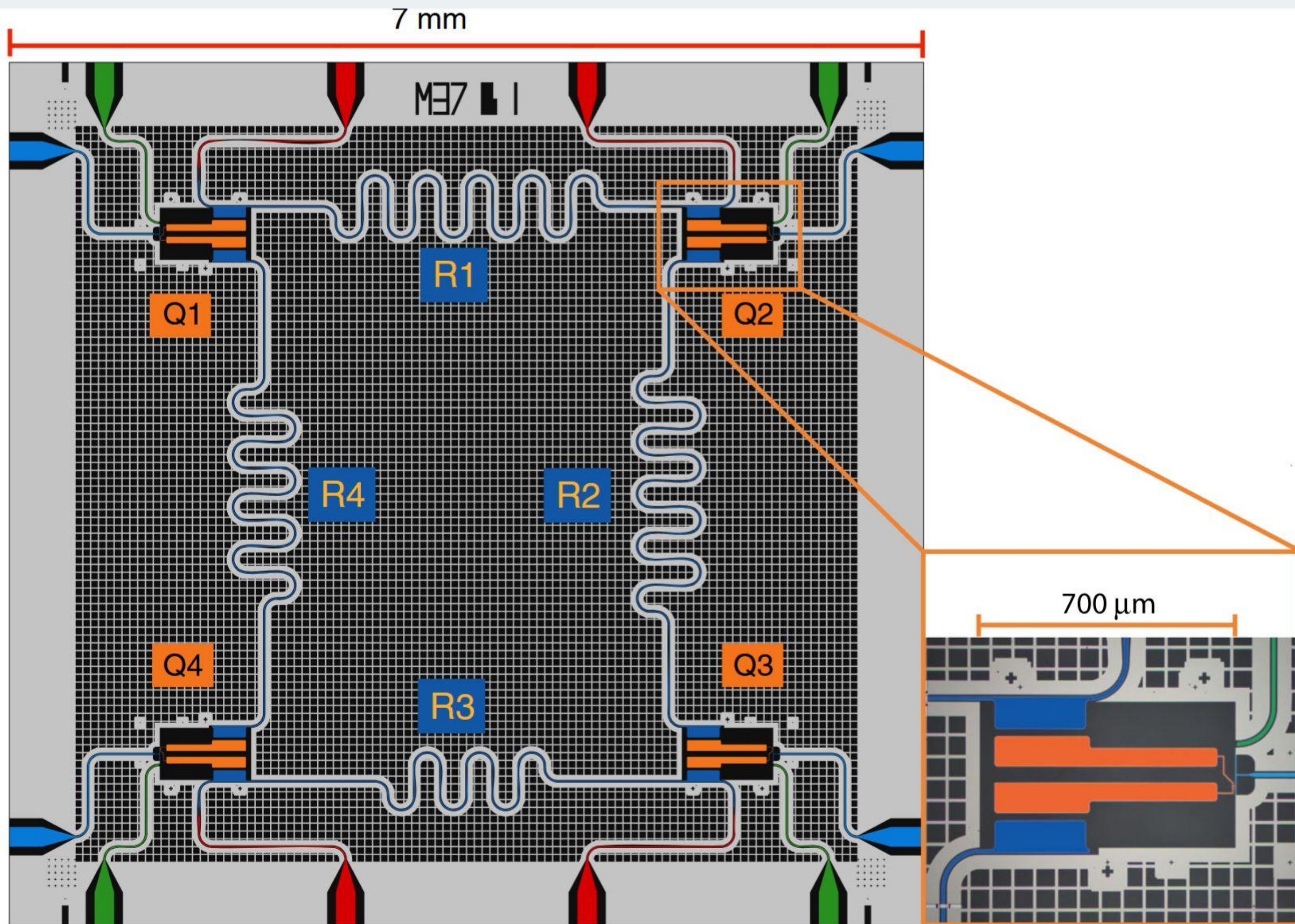


Design and simulation of superconducting circuit

MAXIMILIANO GATTO



1 mm

Q_2

Q_4

R_2

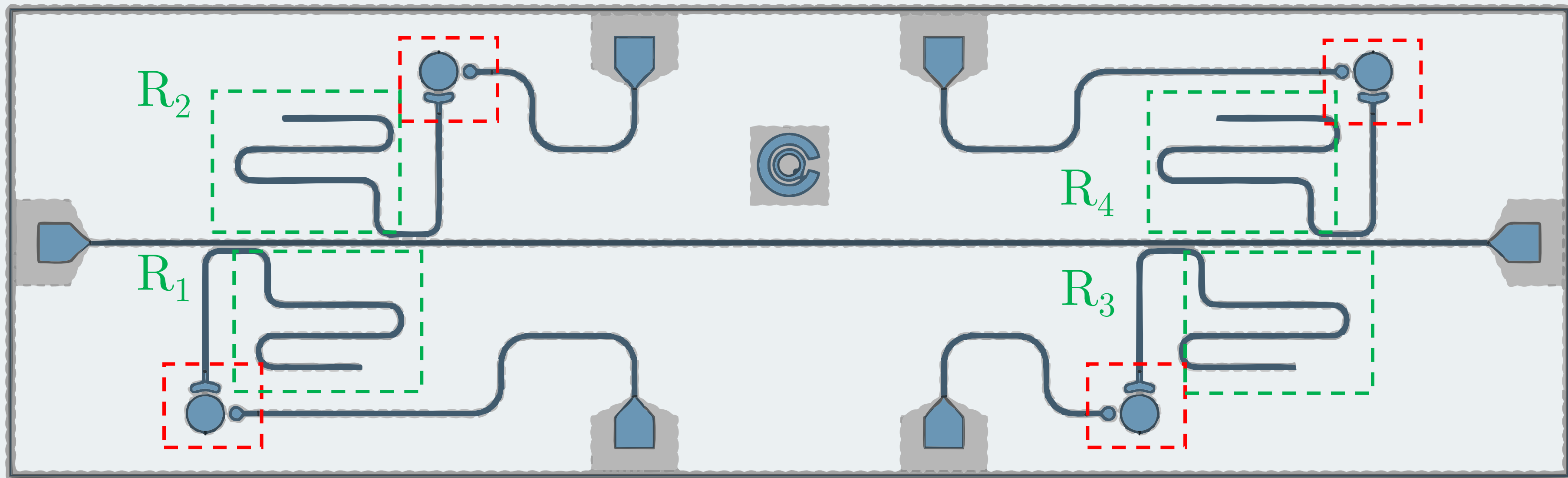
R_4

R_1

R_3

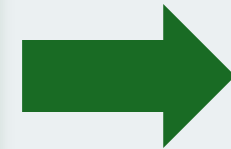
Q_1

Q_3





State of the art



Design CAD



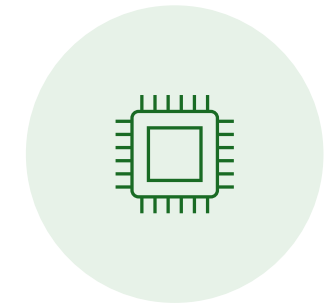
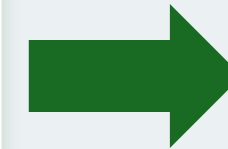
EM simulations



Hamiltonian



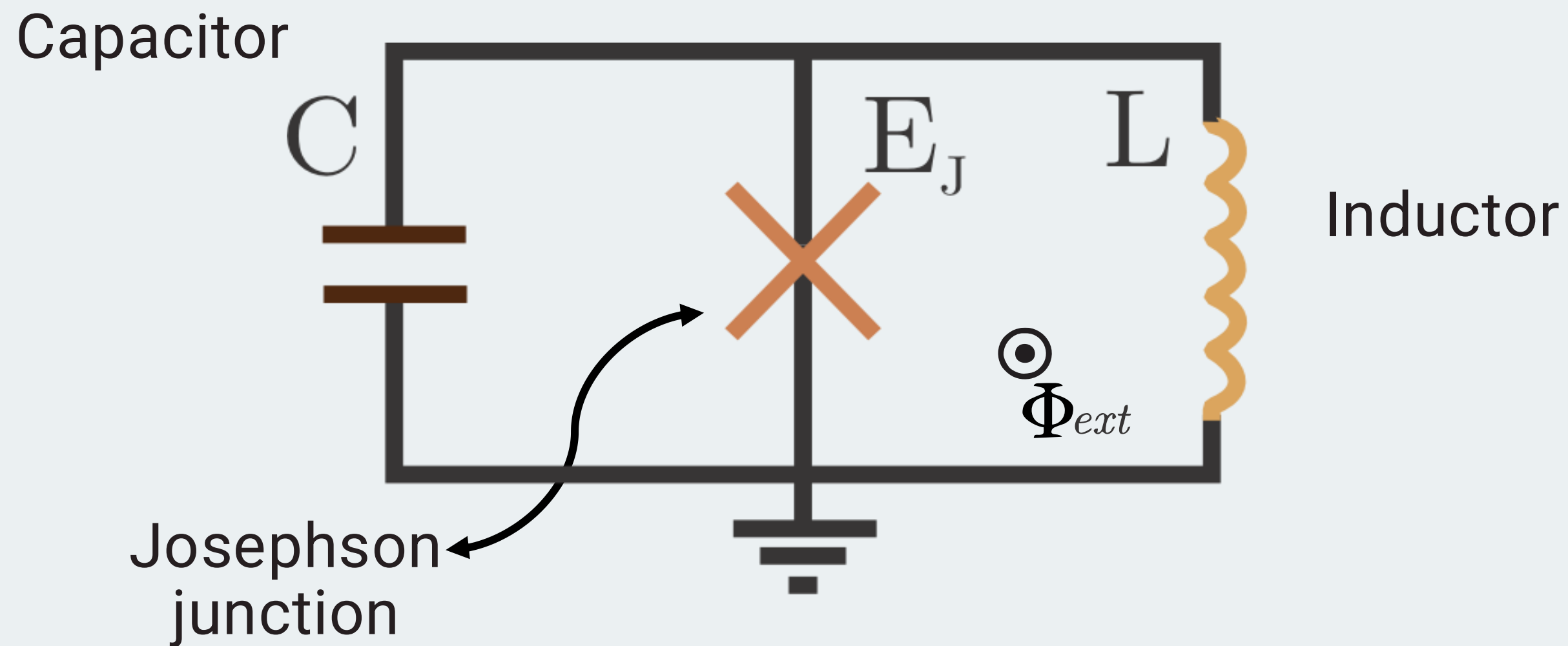
Control



Fabrication

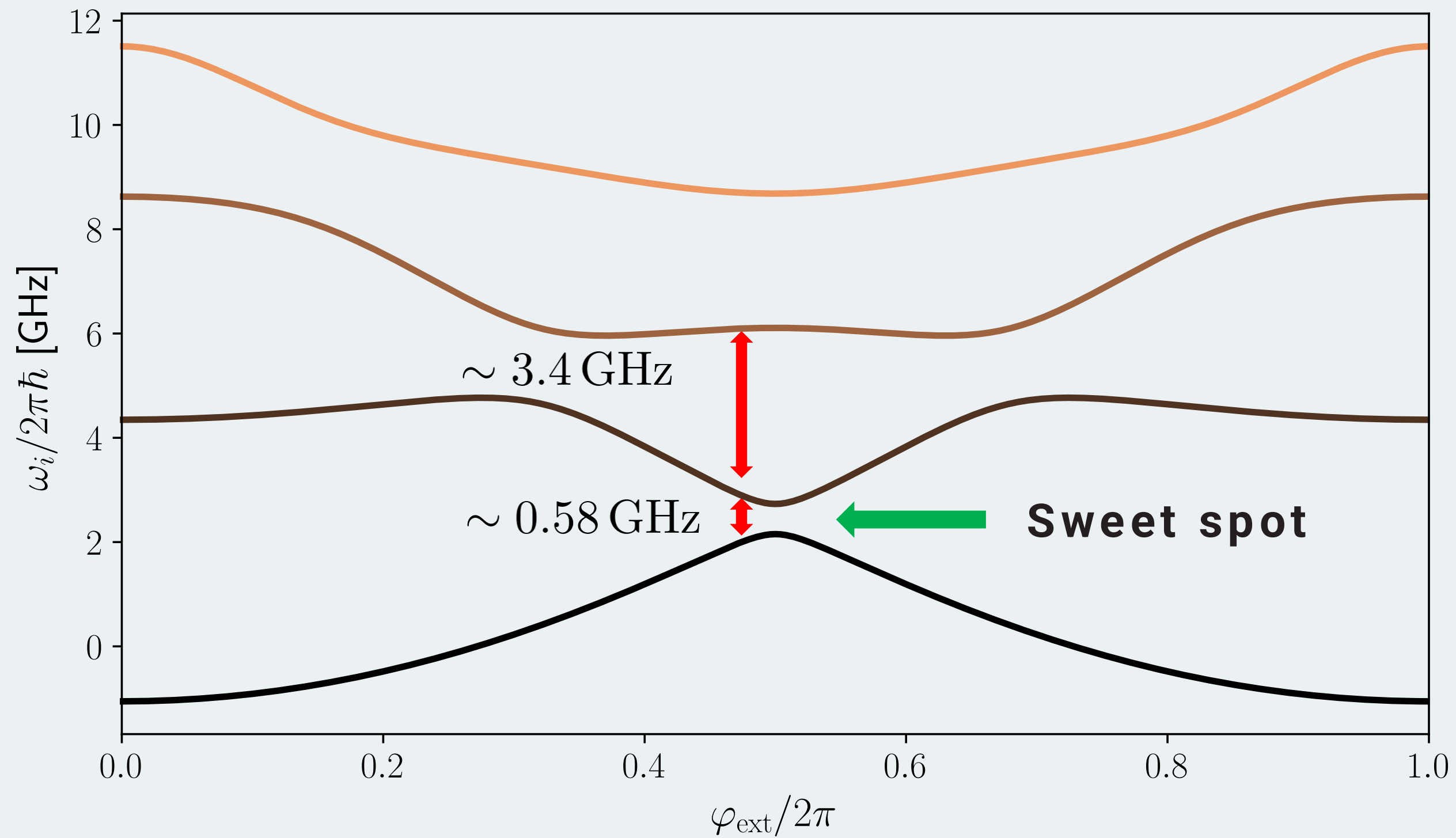
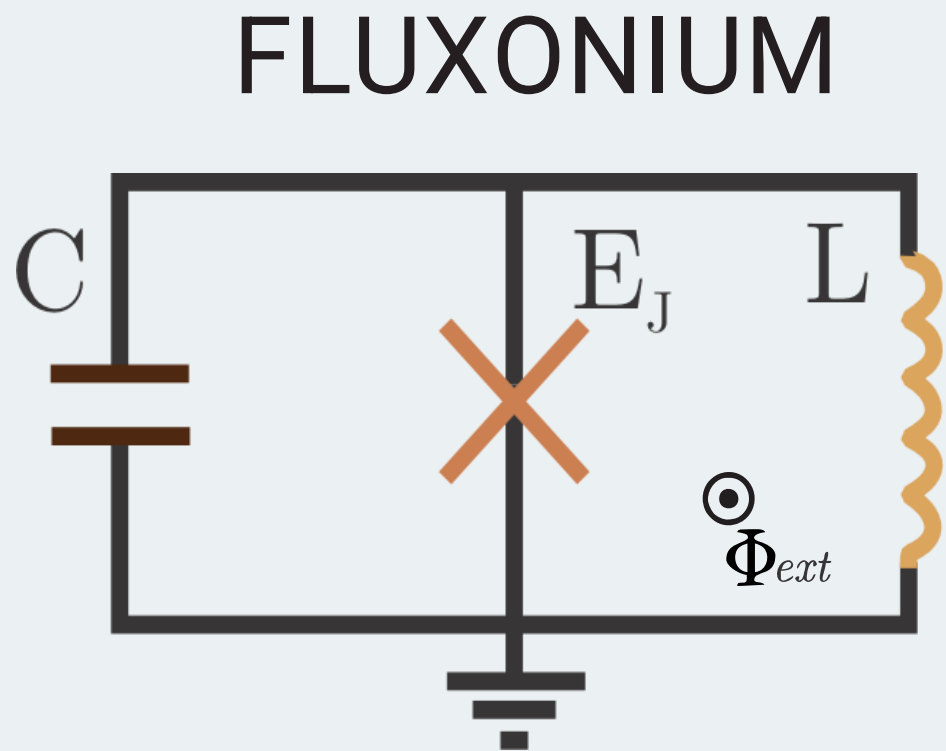
FLUXONIUM QUBIT

FLUXONIUM

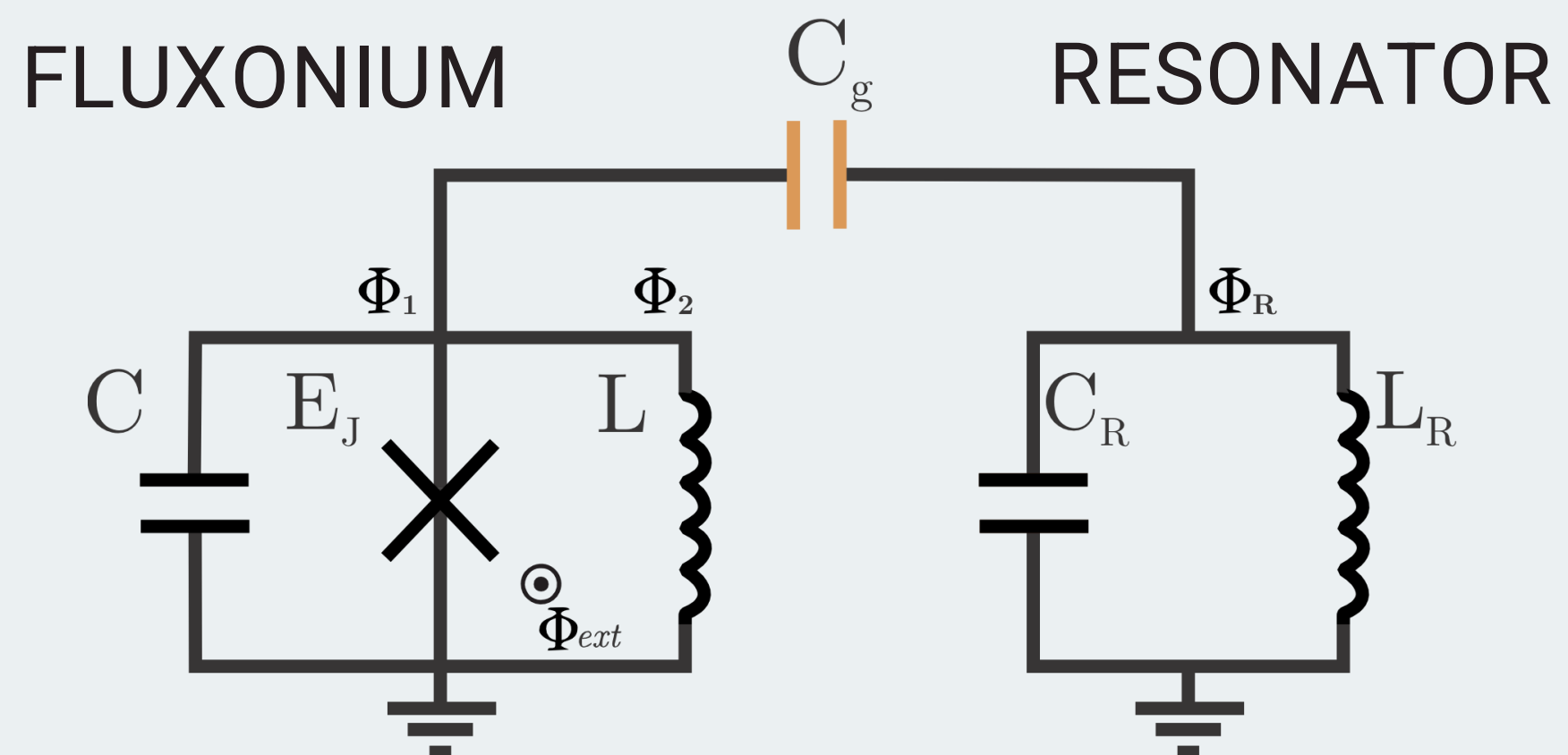


FLUXONIUM QUBIT

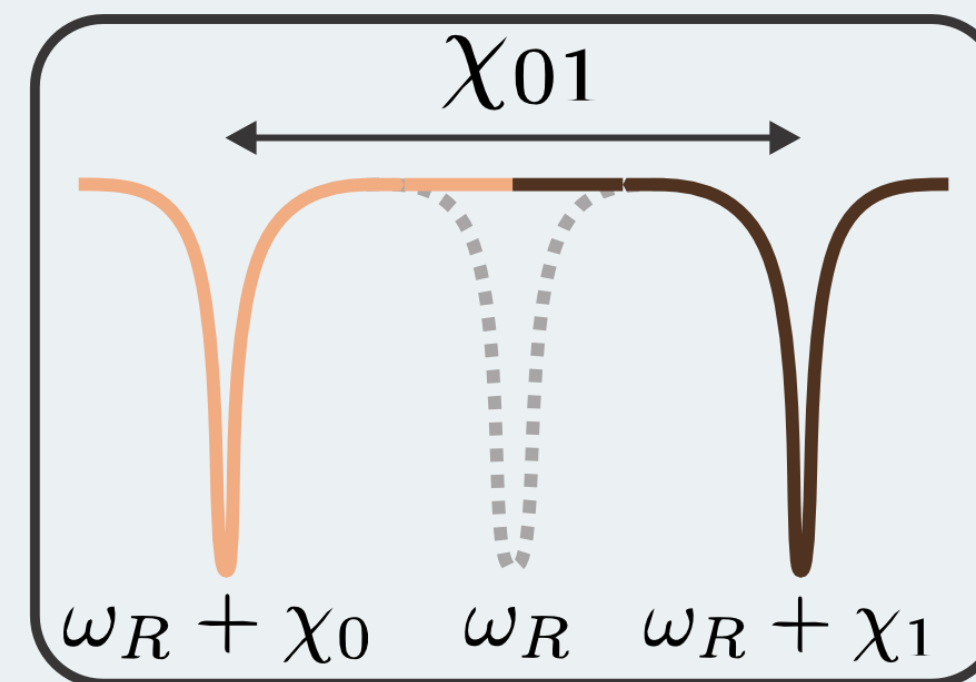
$$\hat{\mathcal{H}} = 4E_C \hat{n}^2 - E_J \cos(\hat{\varphi}) + \frac{E_L}{2} (\hat{\varphi} - \varphi_e)^2 \quad \{E_J, E_C, E_L\} = \{4, 1, 1\} \text{ GHz}$$



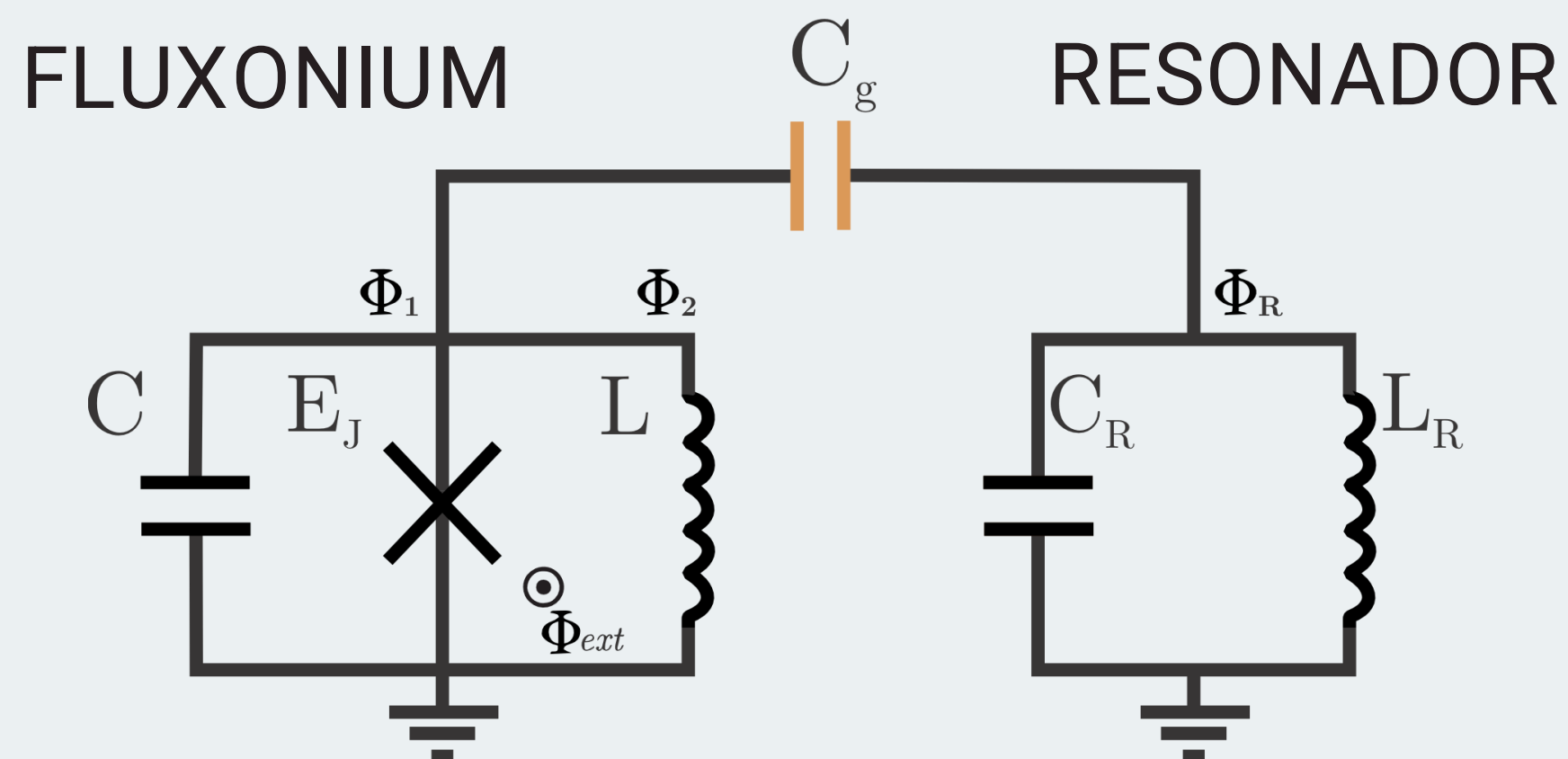
READOUT



$$\omega_R(j) = \omega_R + \chi_j$$

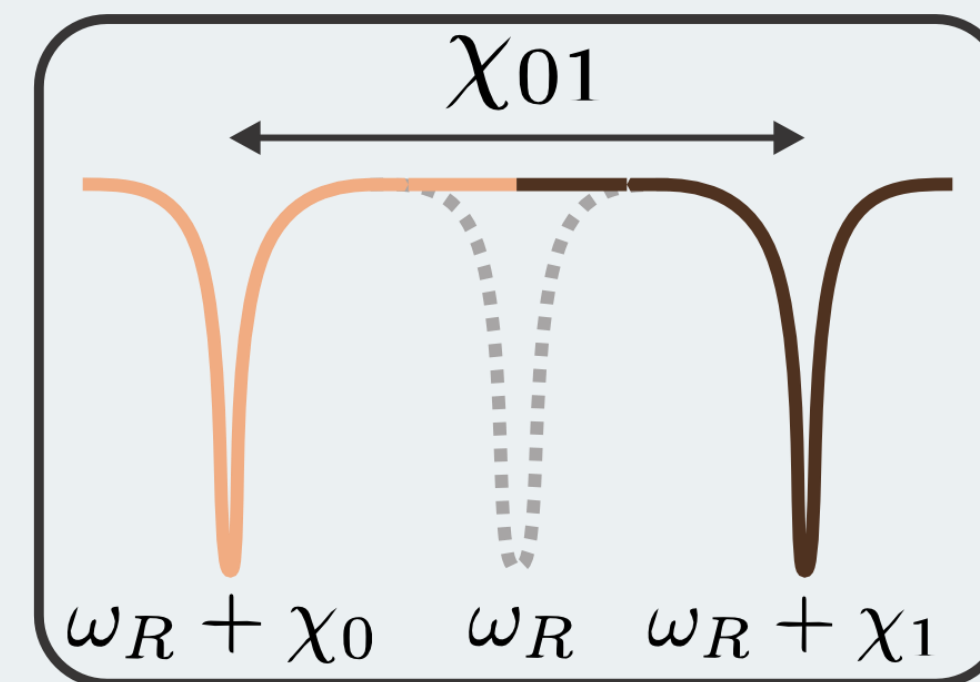


READOUT



$$Q \sim 2500$$

$$\omega_R(j) = \omega_R + \chi_j$$



WEAK COUPLING $C_G \ll C_J, C_R$

Circuit QED

$$\hat{\mathcal{H}} = \underbrace{\left[4E_C \hat{n}^2 + V(\hat{\varphi}) \right]}_{\text{FLUXONIUM (Artificial atom)}} + \underbrace{\hbar\omega_R \hat{a}^\dagger \hat{a}}_{\text{EM FIELD}} - \underbrace{i\hbar g \hat{n} (\hat{a}^\dagger - \hat{a})}_{\text{INT. FLUXONIUM-FIELD}}$$

IMPLEMENTATION

¿Qué necesitamos para construir un fluxonium?

$$\{E_J, E_C, E_L\} = \{4, 1, 1\} \text{ GHz}$$

$$E_C = \frac{e^2}{2C} \longrightarrow C \sim 20 \text{ fF}$$

$$E_J = I_0 \varphi_0 \longrightarrow I_0 \sim 8 \text{ nA}$$

$$E_L = \frac{\varphi_0^2}{L} \longrightarrow L \sim 165 \text{ nH}$$

IMPLEMENTATION

¿Qué necesitamos para construir un fluxonium?

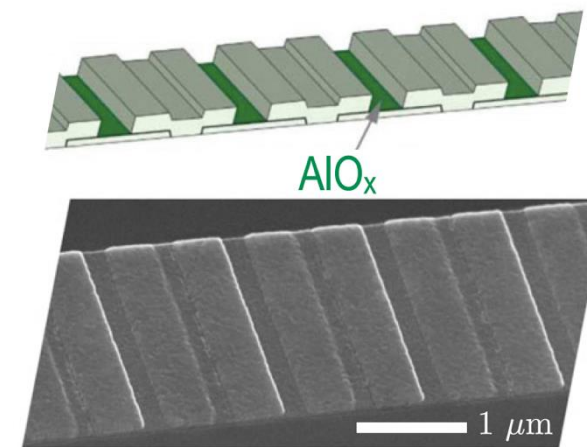
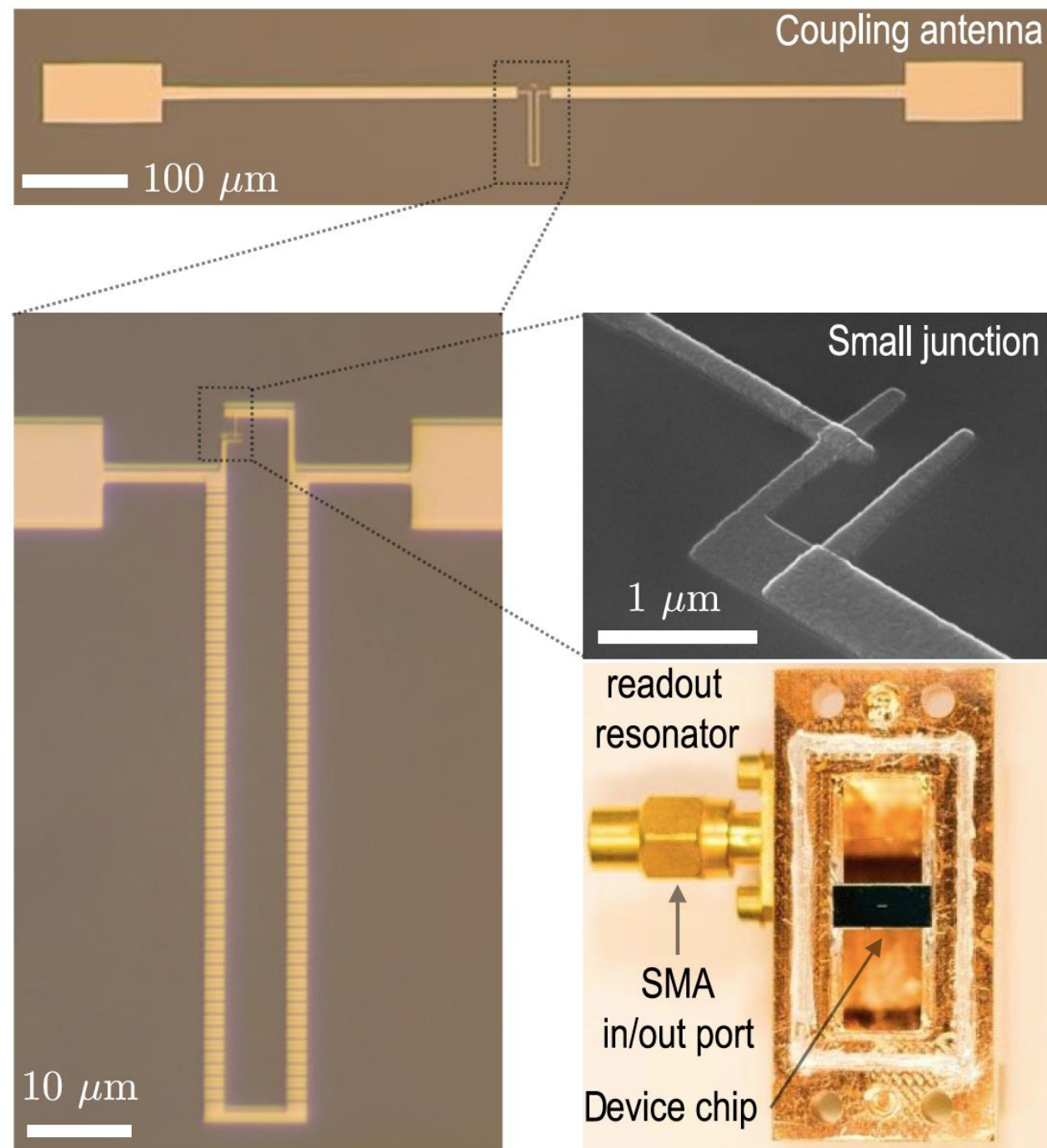
$$\{E_J, E_C, E_L\} = \{4, 1, 1\} \text{ GHz}$$

$$E_L = \frac{\varphi_0^2}{L} \longrightarrow L \sim 165 \text{ nH}$$

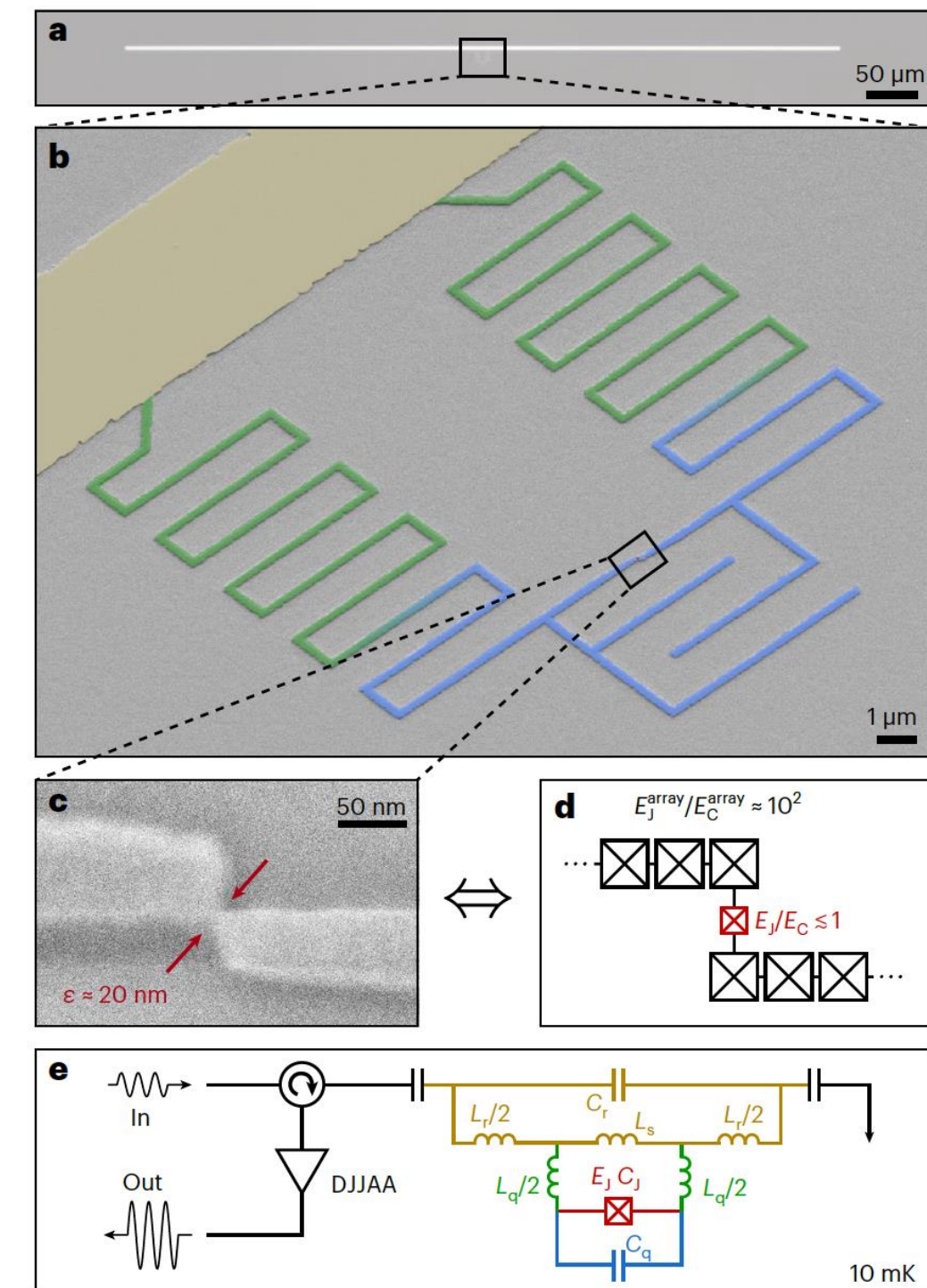
$$L \sim \mu_0 \ell \longrightarrow \boxed{\ell \sim 130 \text{ mm}}$$

$$\mu_0 = 4\pi \times 10^{-7} \frac{\text{H}}{\text{m}}$$

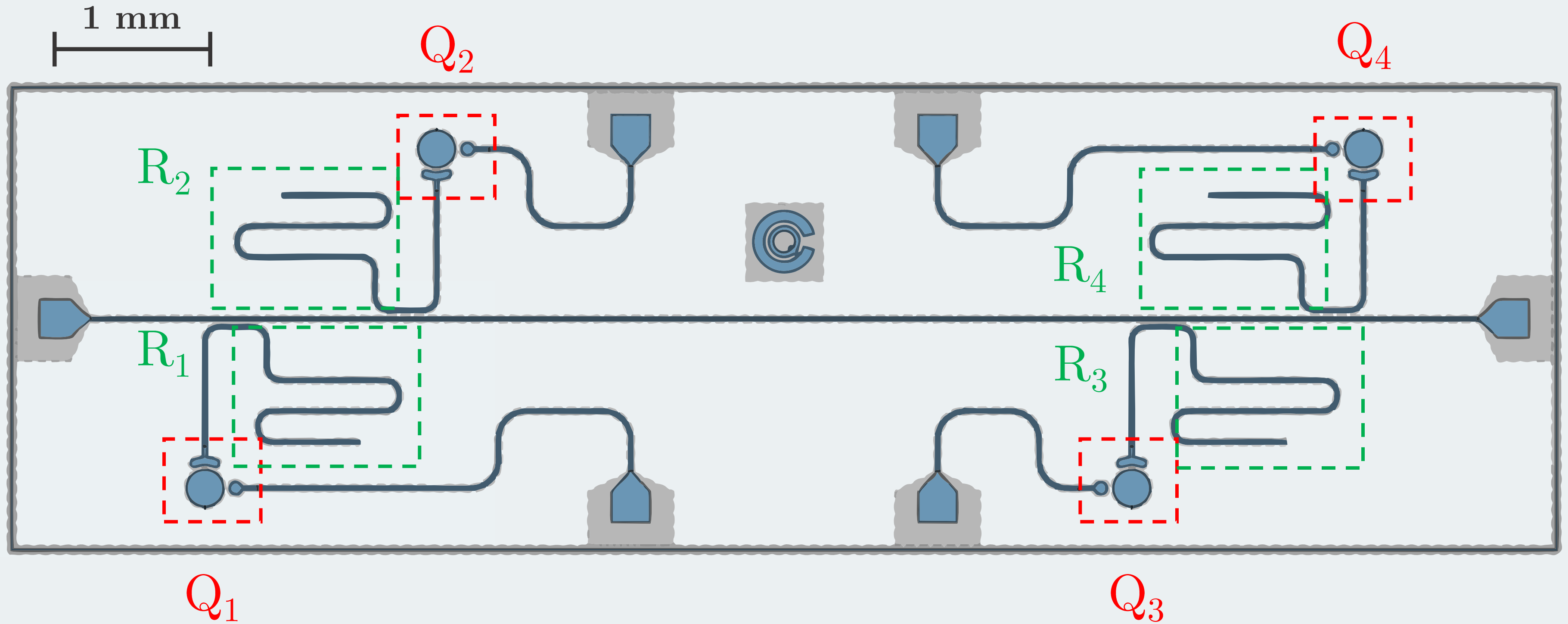
OVERVIEW OF RECENT PROGRESS



GRALMONIUM

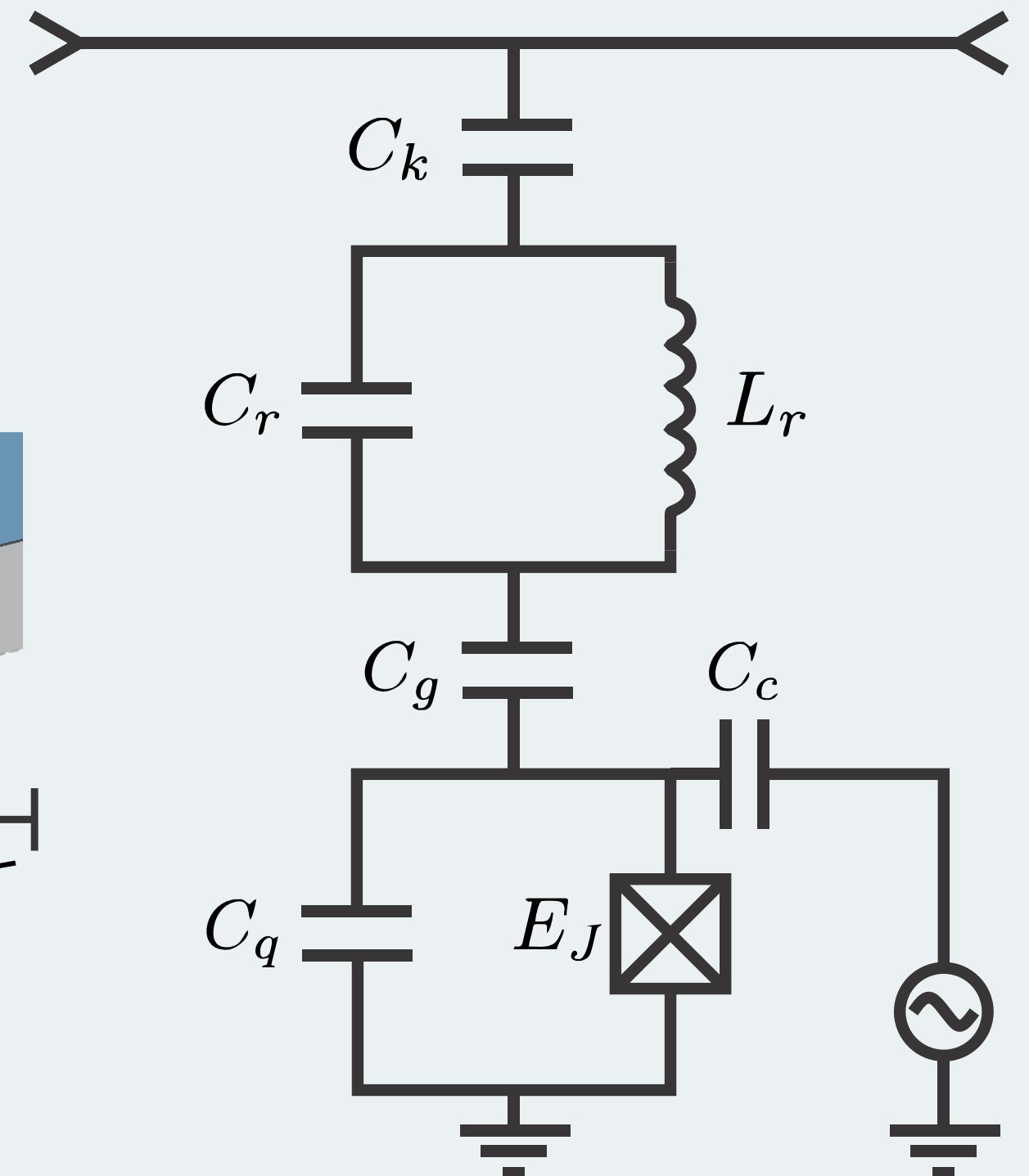
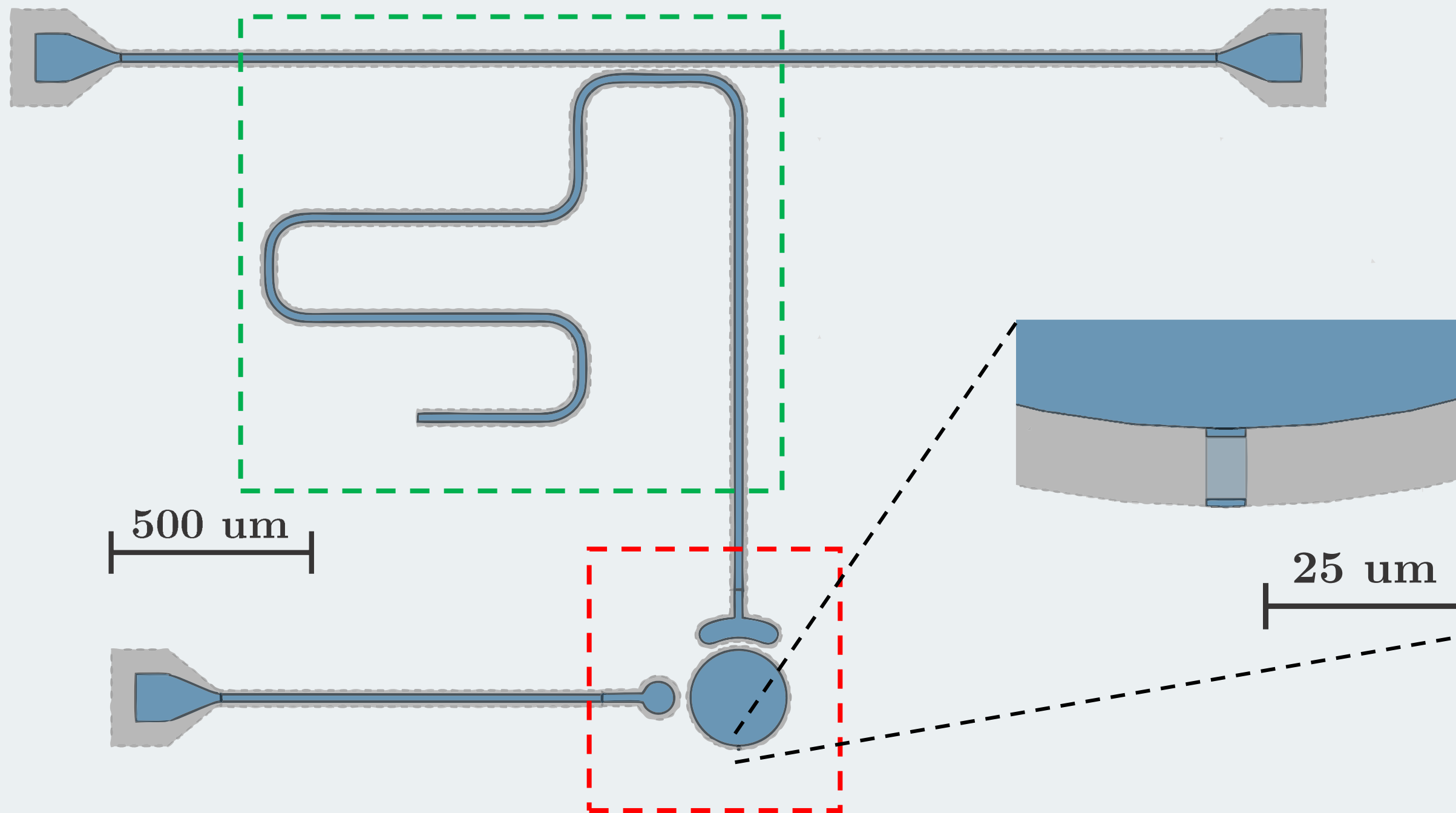
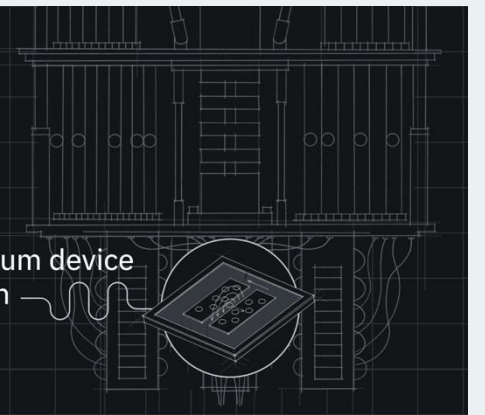


DESIGN OF 4 TRANSMON CHIP

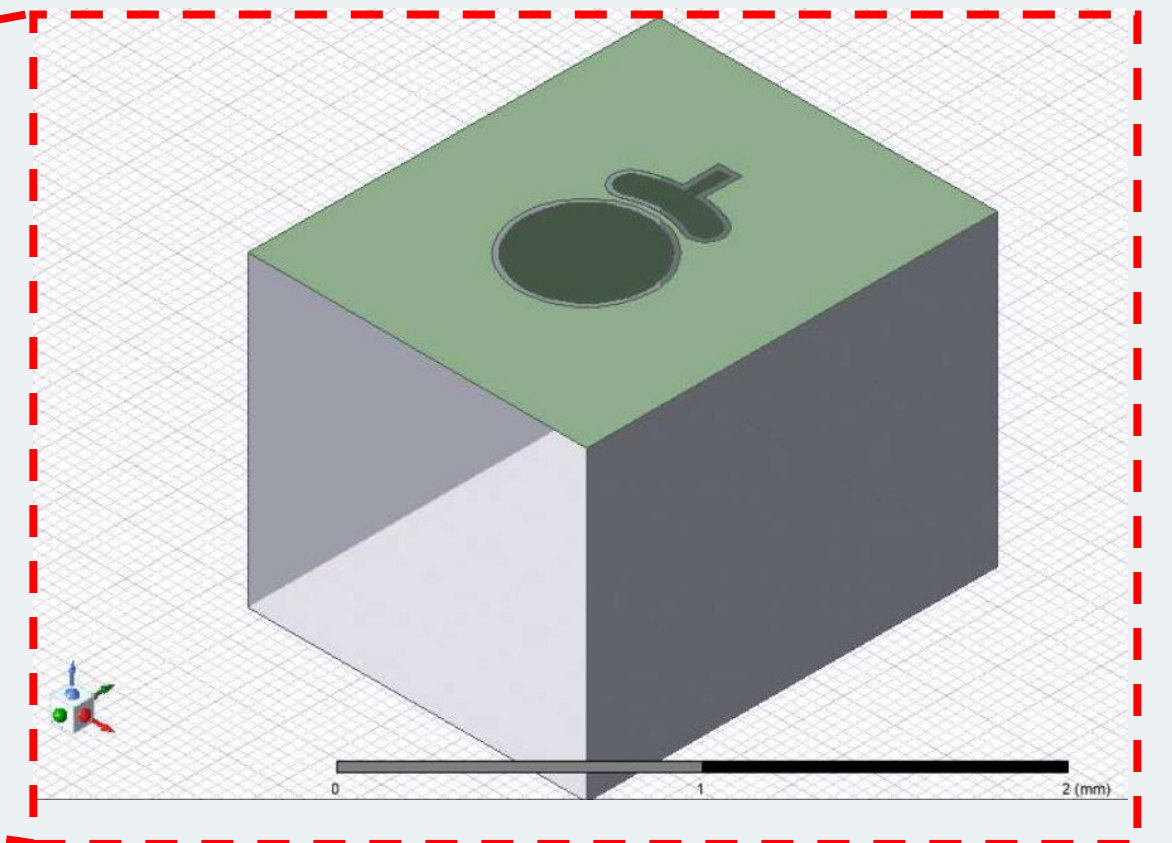
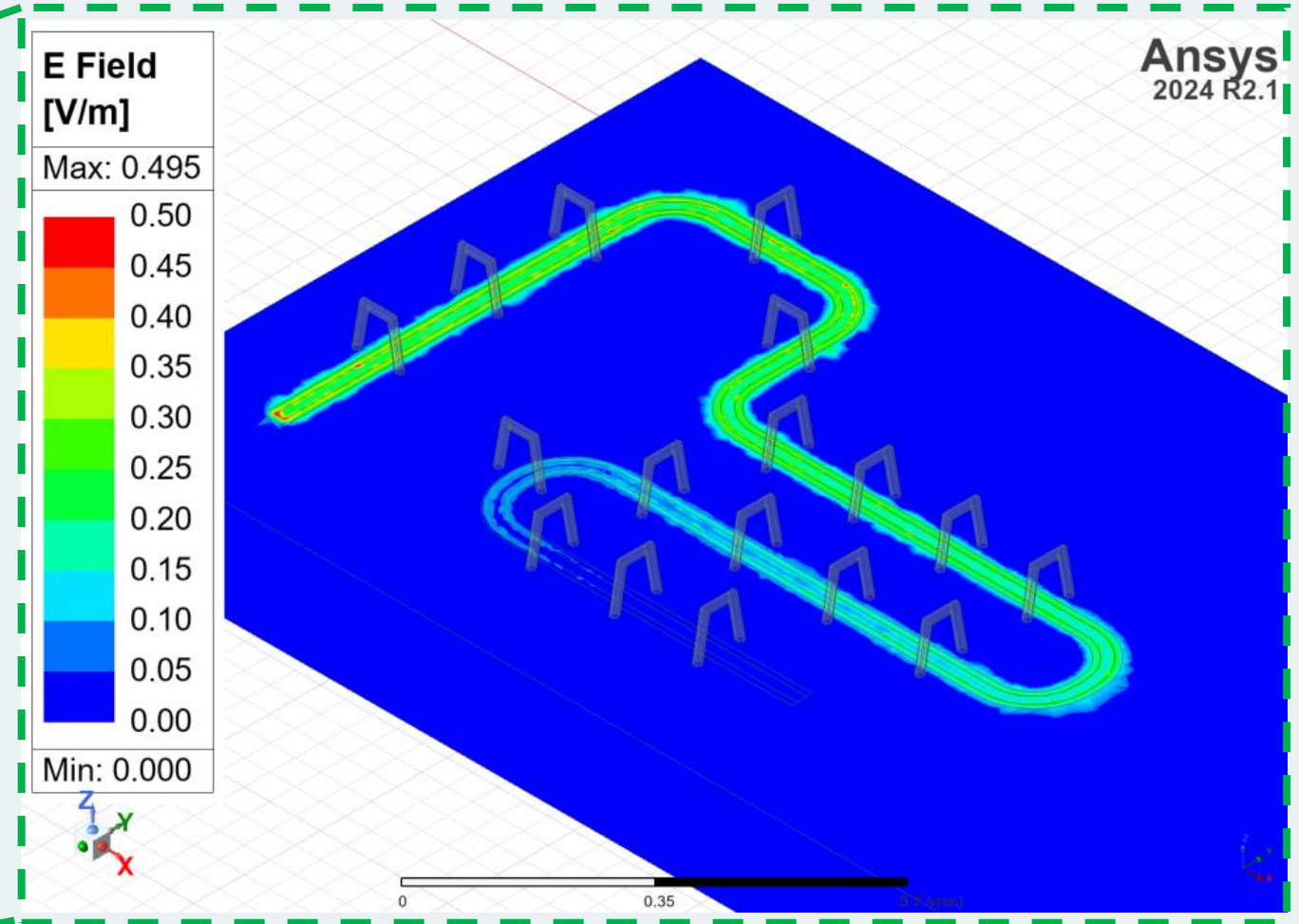
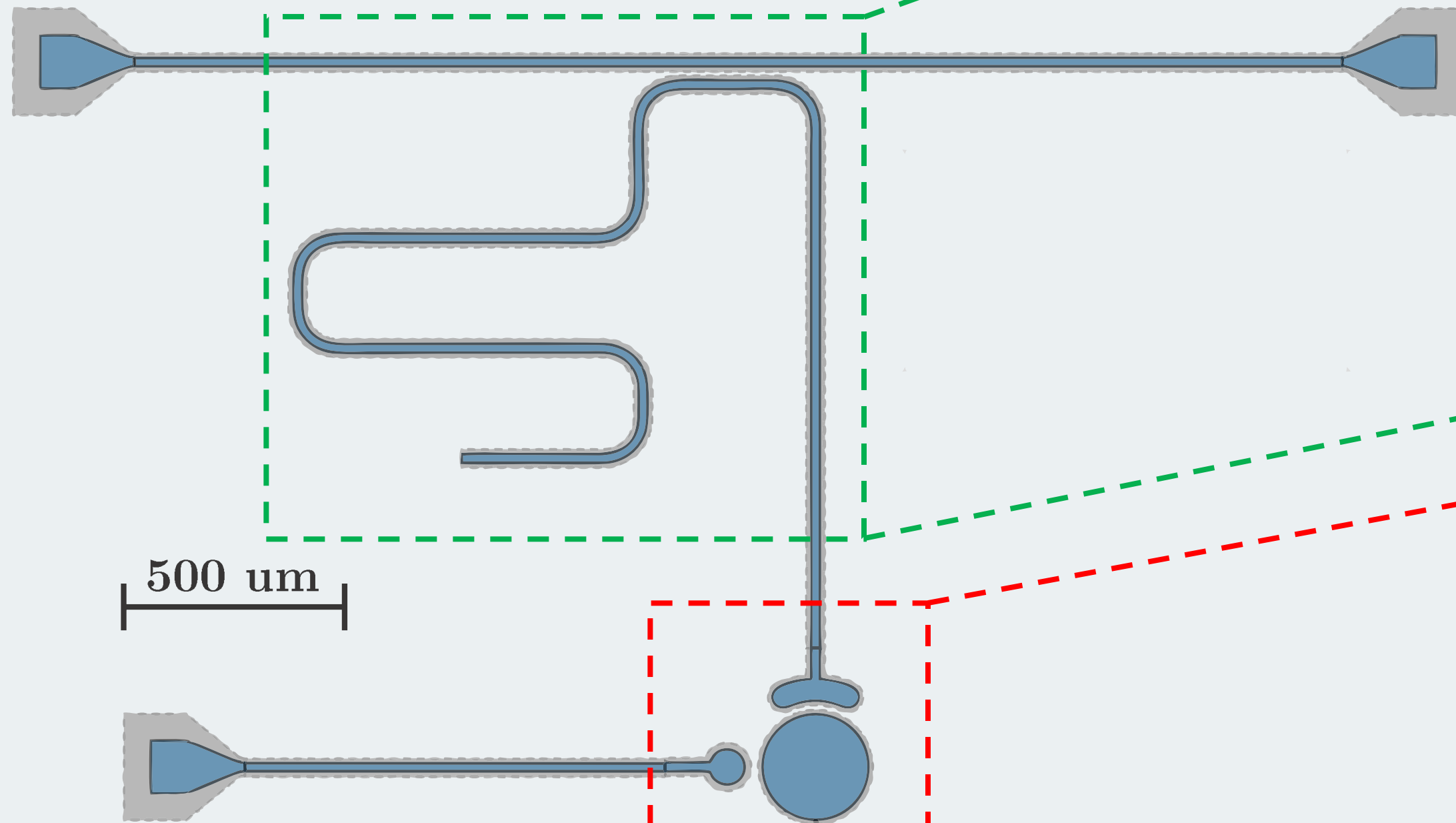


EJEMPLO TRANSMON

Qiskit | quantum device design



EJEMPLO TRANSMON



Open-Source Highly Parallel Electromagnetic Simulations for Superconducting Circuits

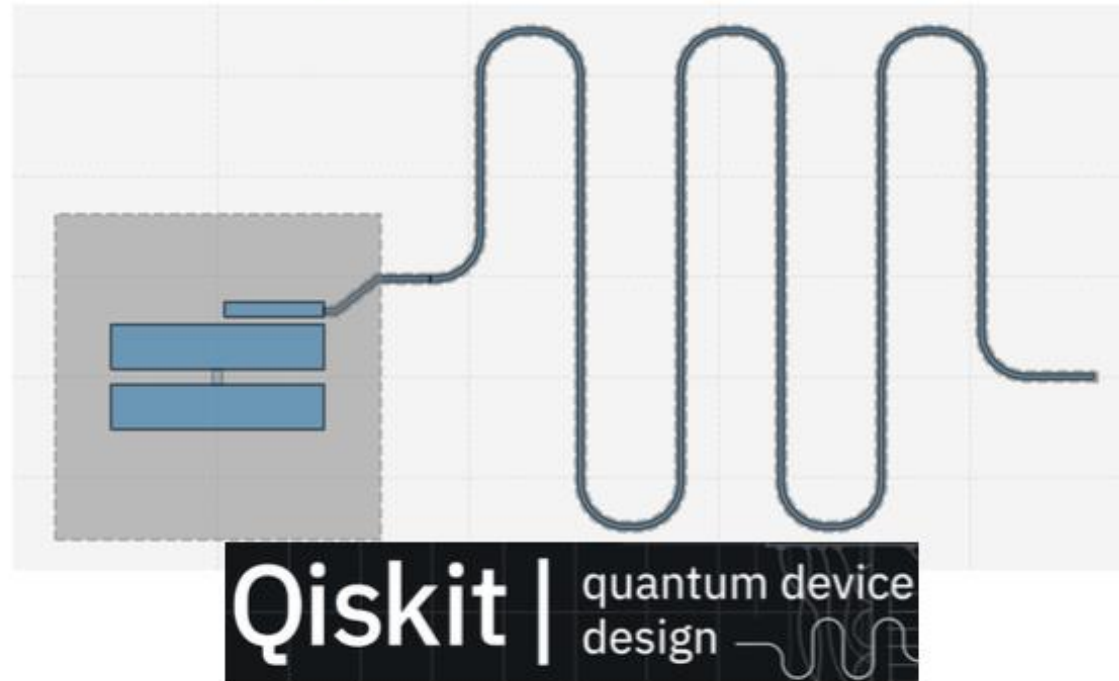
David Sommers, Zach Degnan, Divita Gautam, Yi-Hsun Chen,
Chun-Ching Chiu, Arkady Fedorov, and Prasanna Pakkiam*

*School of Mathematics and Physics,
University of Queensland, Brisbane, Australia*

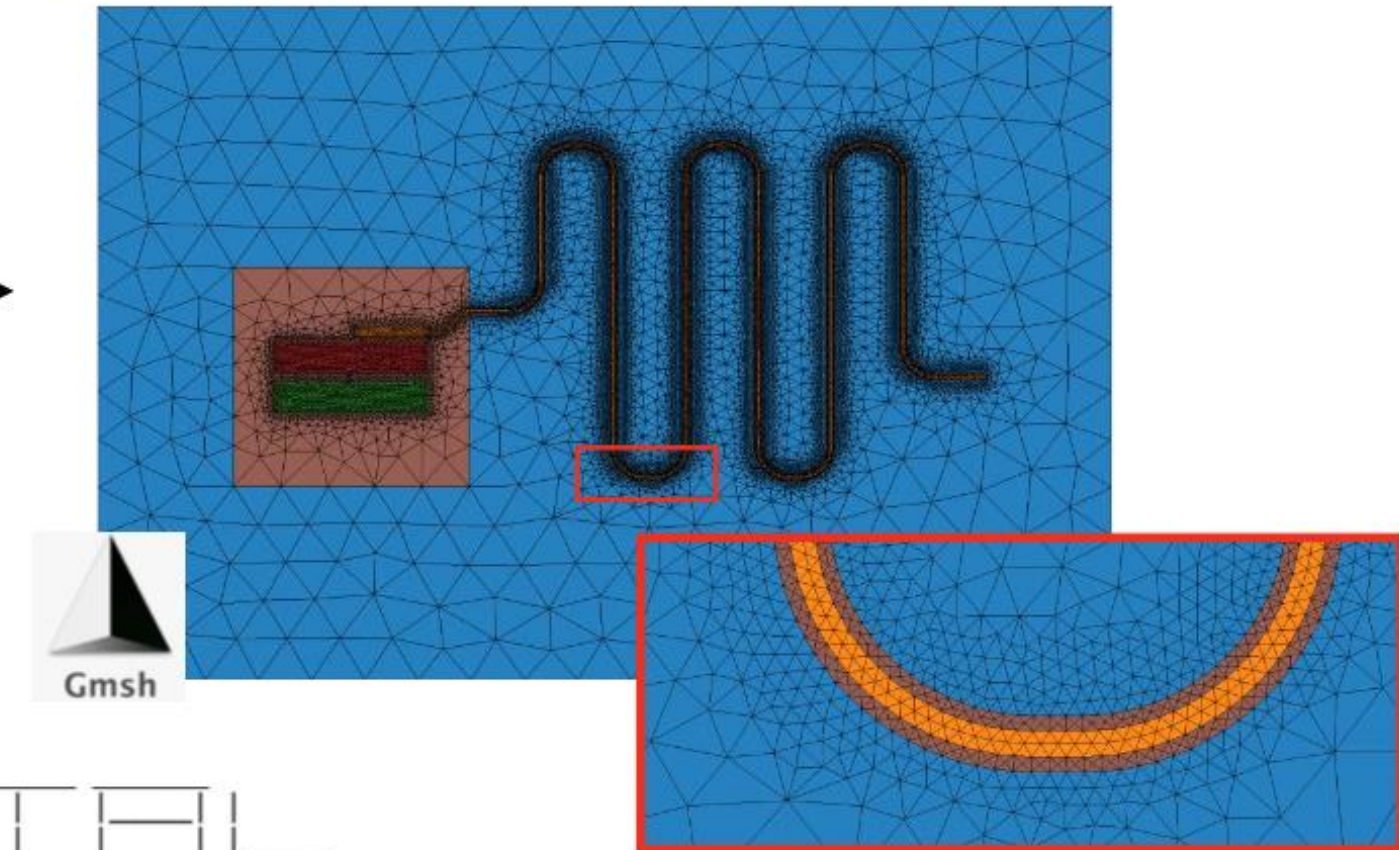
(Dated: November 7, 2025)

Electromagnetic simulations form an indispensable part of the design and optimization process for superconducting quantum devices. Although several commercial platforms exist, open-source alternatives optimized for high-performance computing remain limited. To address this gap, we introduce SQDMetal, a Python-based API that integrates Qiskit Metal (IBM), Gmsh, Palace (AWS), and Paraview (Kitware) into an open-source, highly parallel simulation workflow for superconducting quantum circuits. SQDMetal enables accurate, efficient, and scalable simulations while remaining community-driven and free from commercial constraints. In this work, we validate SQDMetal through mesh convergence studies which benchmark SQDMetal against COMSOL Multiphysics and Ansys, demonstrating excellent agreement for both eigenmode and electrostatic (capacitance) simulations. Furthermore, we simulate superconducting resonators and transmon qubits, showing reasonable agreement with experimental measurements. SQDMetal also supports advanced capabilities, including Hamiltonian extraction via the energy participation ratio (EPR) method, incorporation of kinetic inductance effects, and full 3D modeling of device geometry for improved predictive accuracy. By unifying open-source tools into a single framework, SQDMetal lowers the barriers to entry for community members seeking to access high-performance simulations to assist in the design and optimization of their devices.

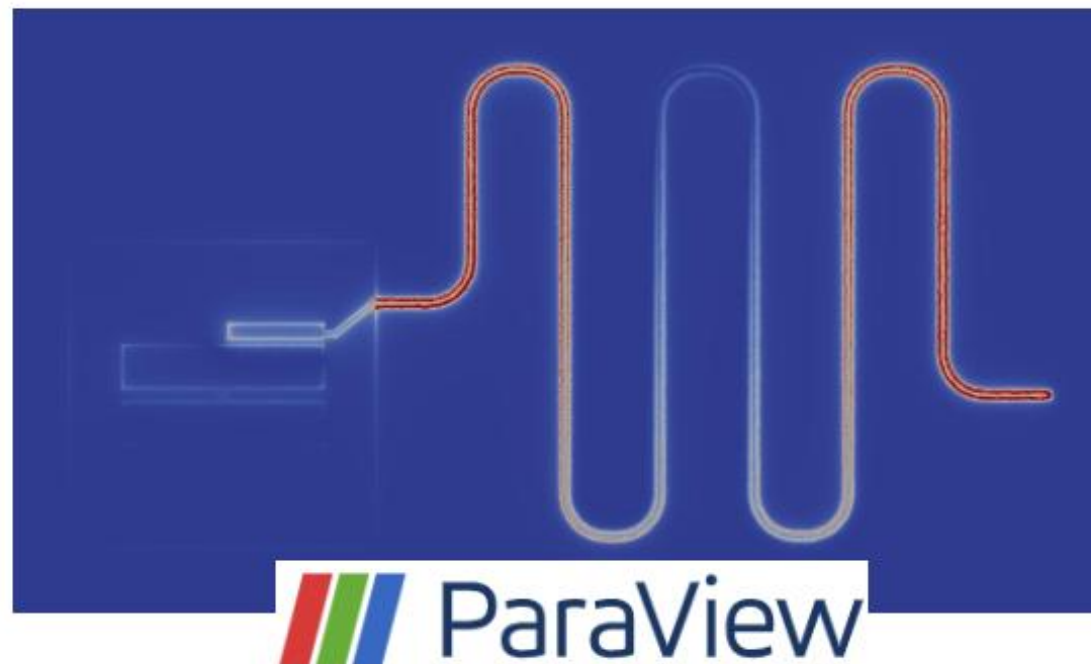
1). Create/Alter Design



2). Mesh Design



4). Visualise Fields

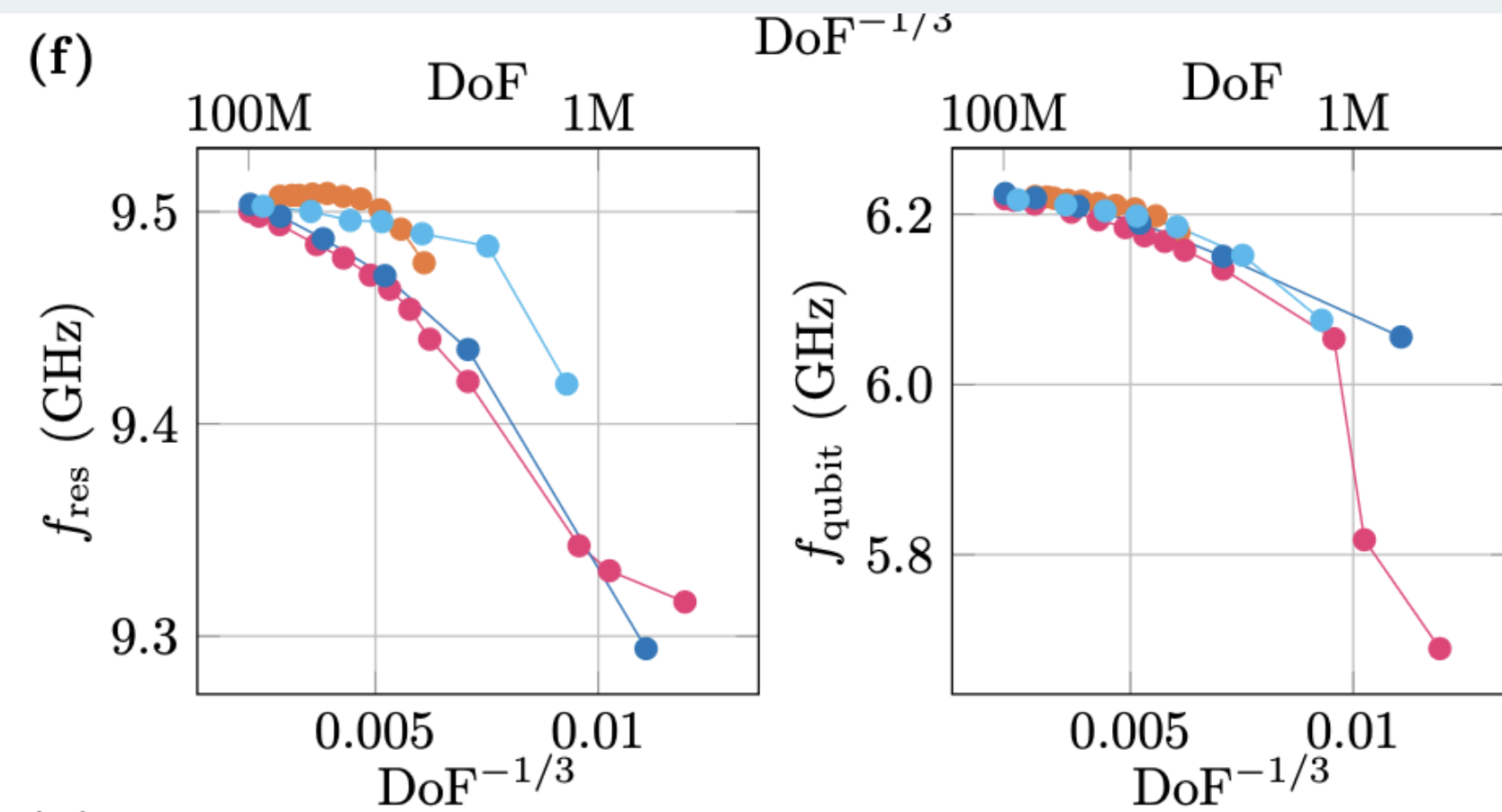
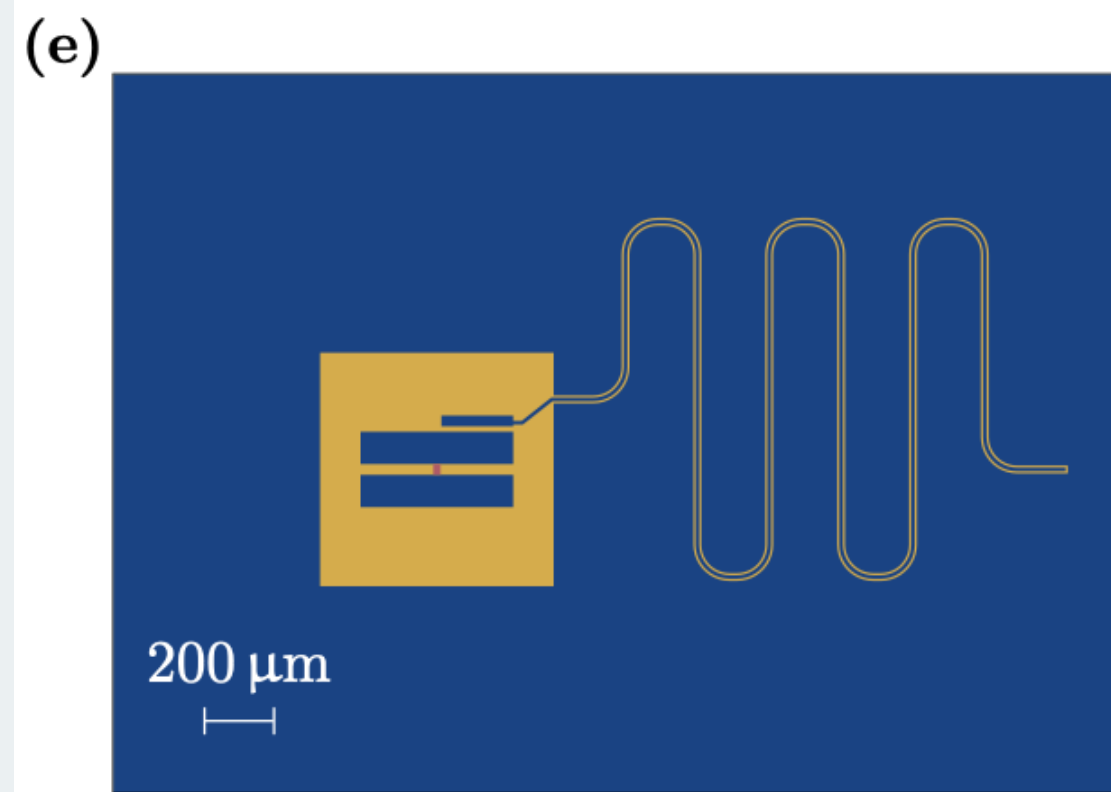
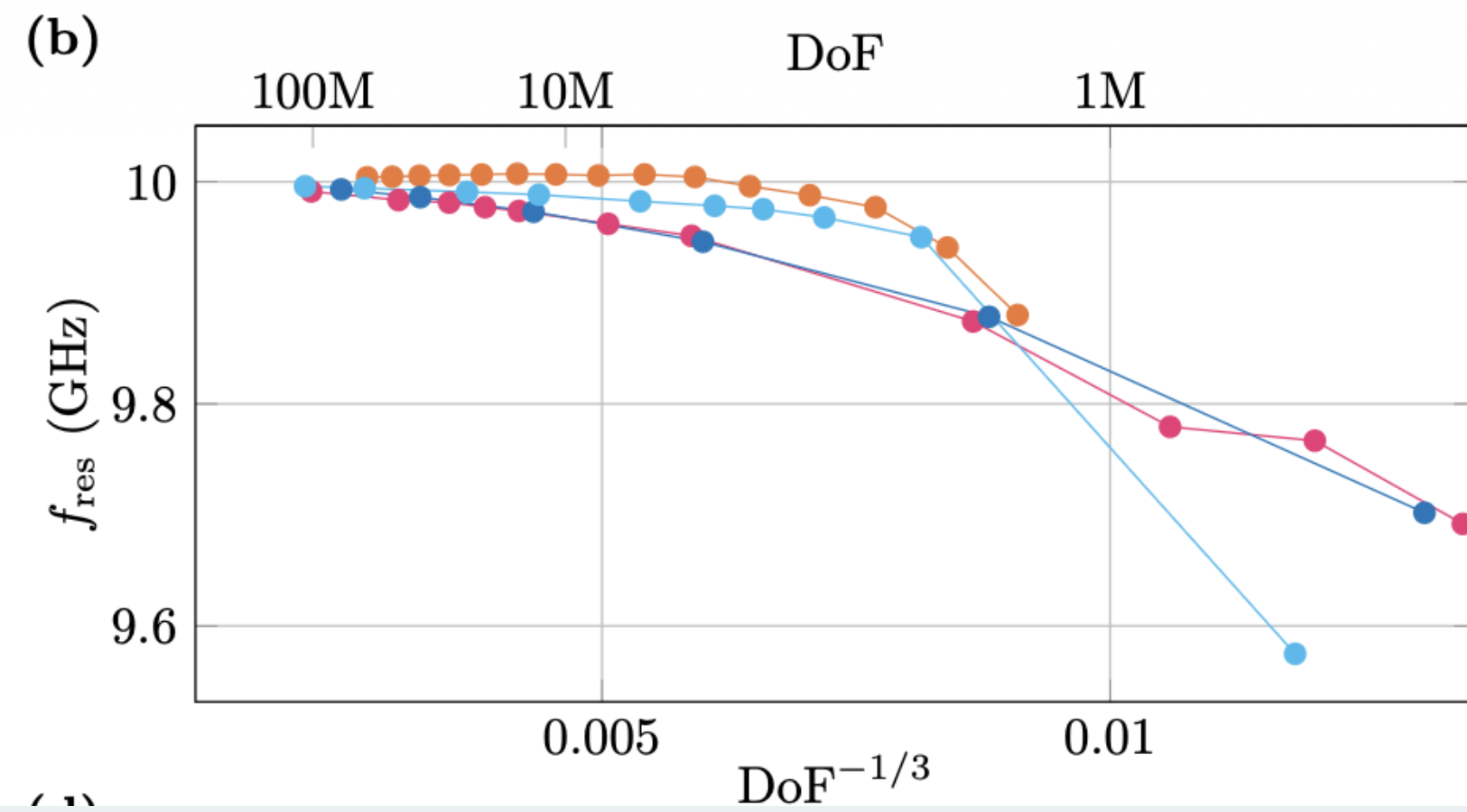
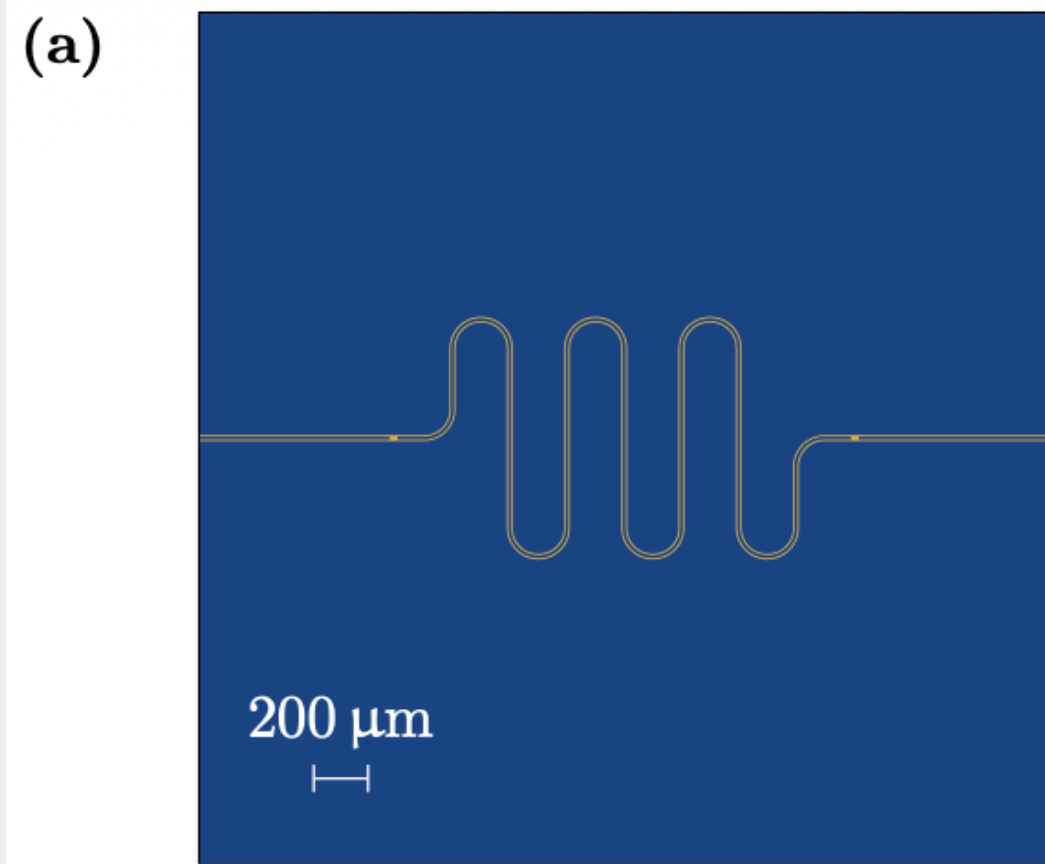


SQUIDMETAL

3). Simulate Design

m,	Re{f} (GHz),	Im{f} (GHz),	Q,
1.000000000e+00,	+6.217032342e+00,	+7.719077005e-06,	+4.027056822e+05,
2.000000000e+00,	+9.439116481e+00,	+1.163134703e-05,	+4.057619661e+05,



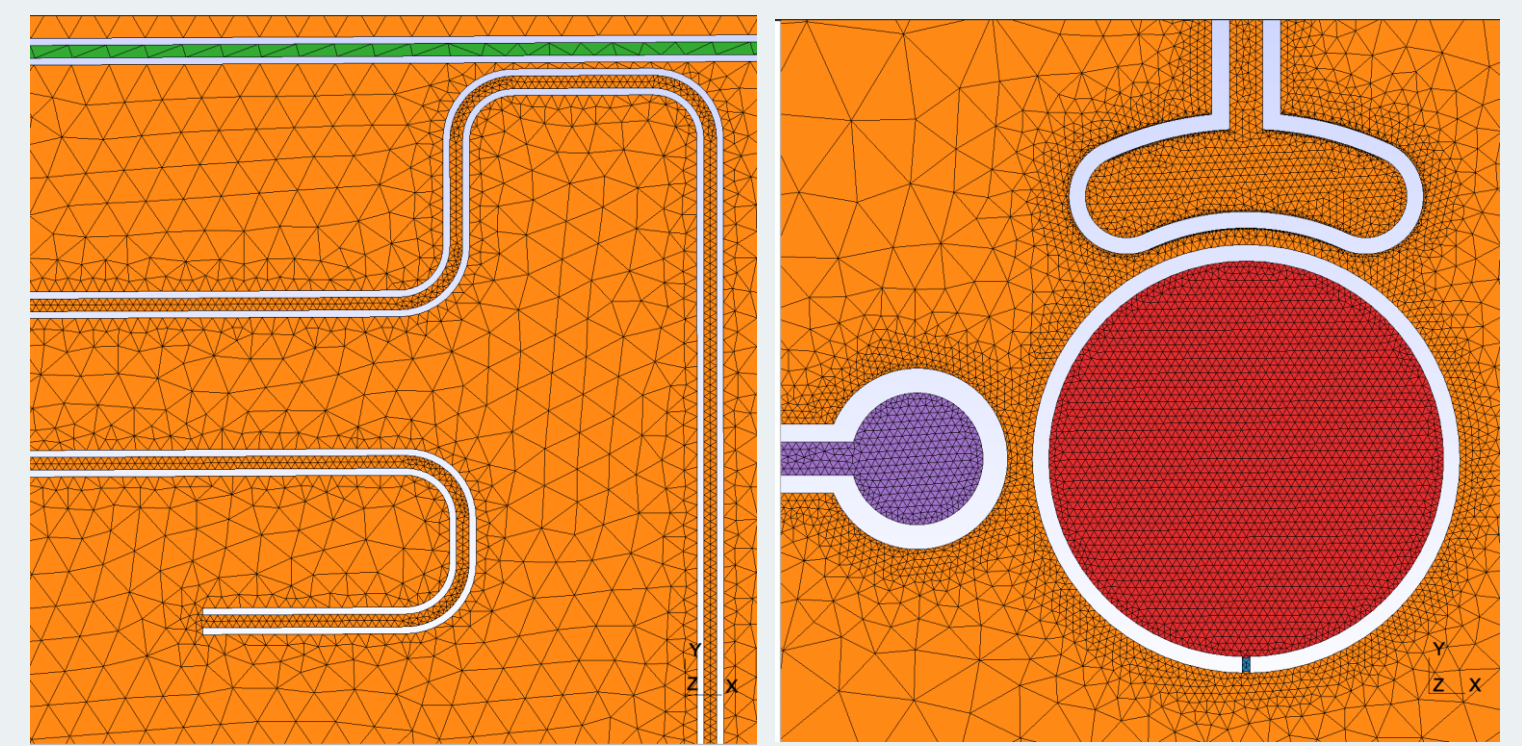
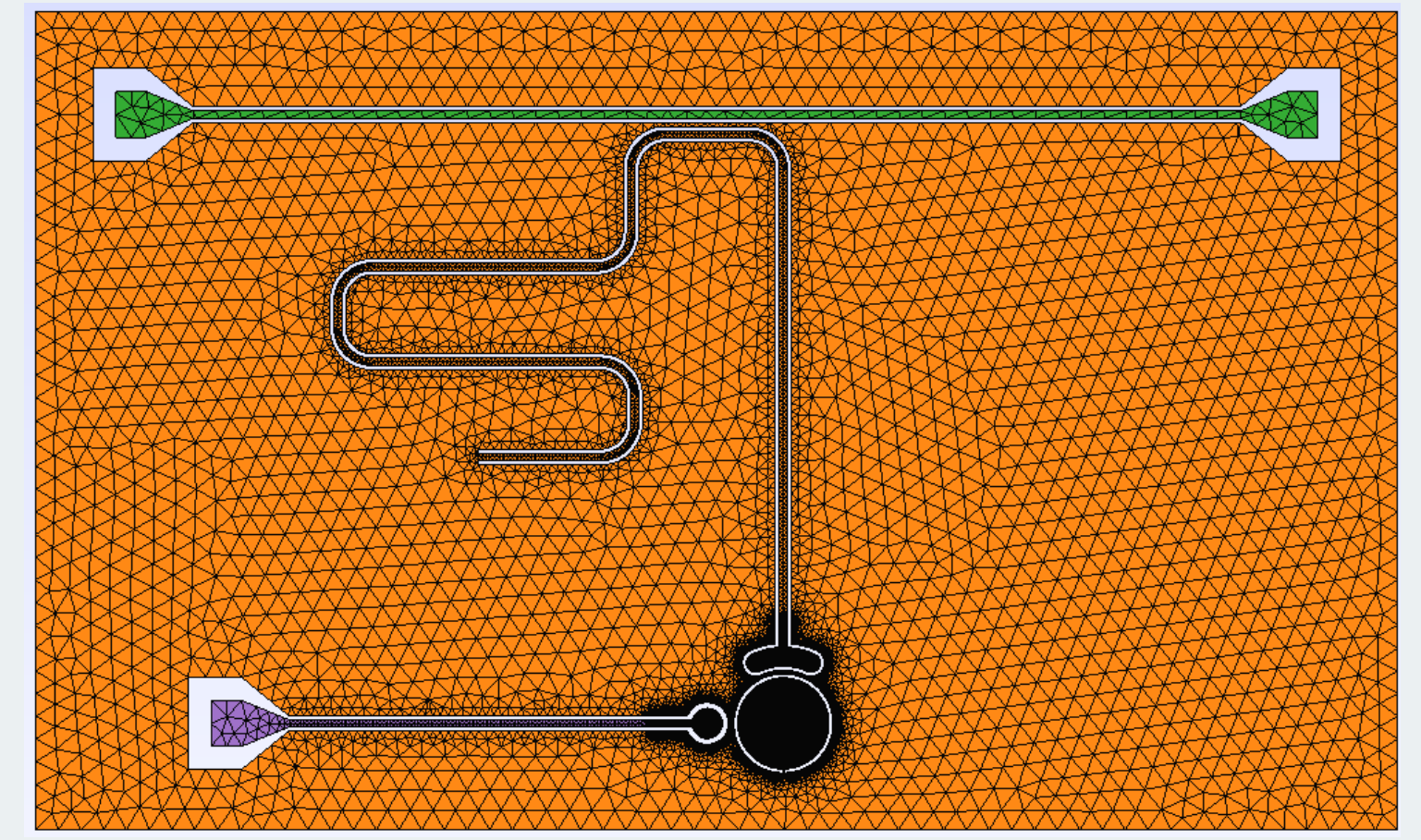
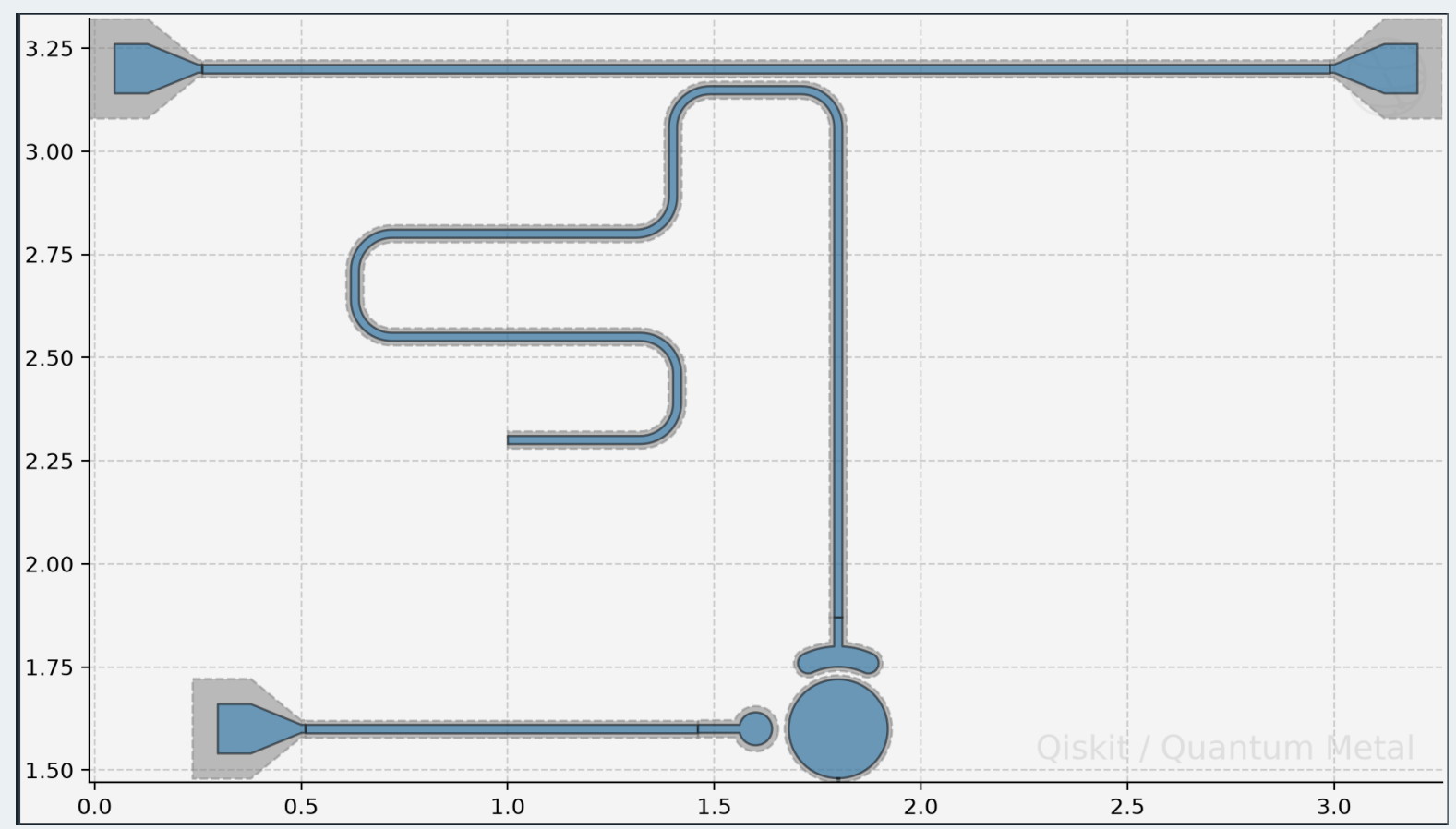


■ Metals (PEC) ■ Dielectric (Si)
■ Josephson Junction

—○— ANSYS (O2/AMR) —●— COMSOL (O2)
—●— Palace (O2/AMR) —●— Palace (O4)

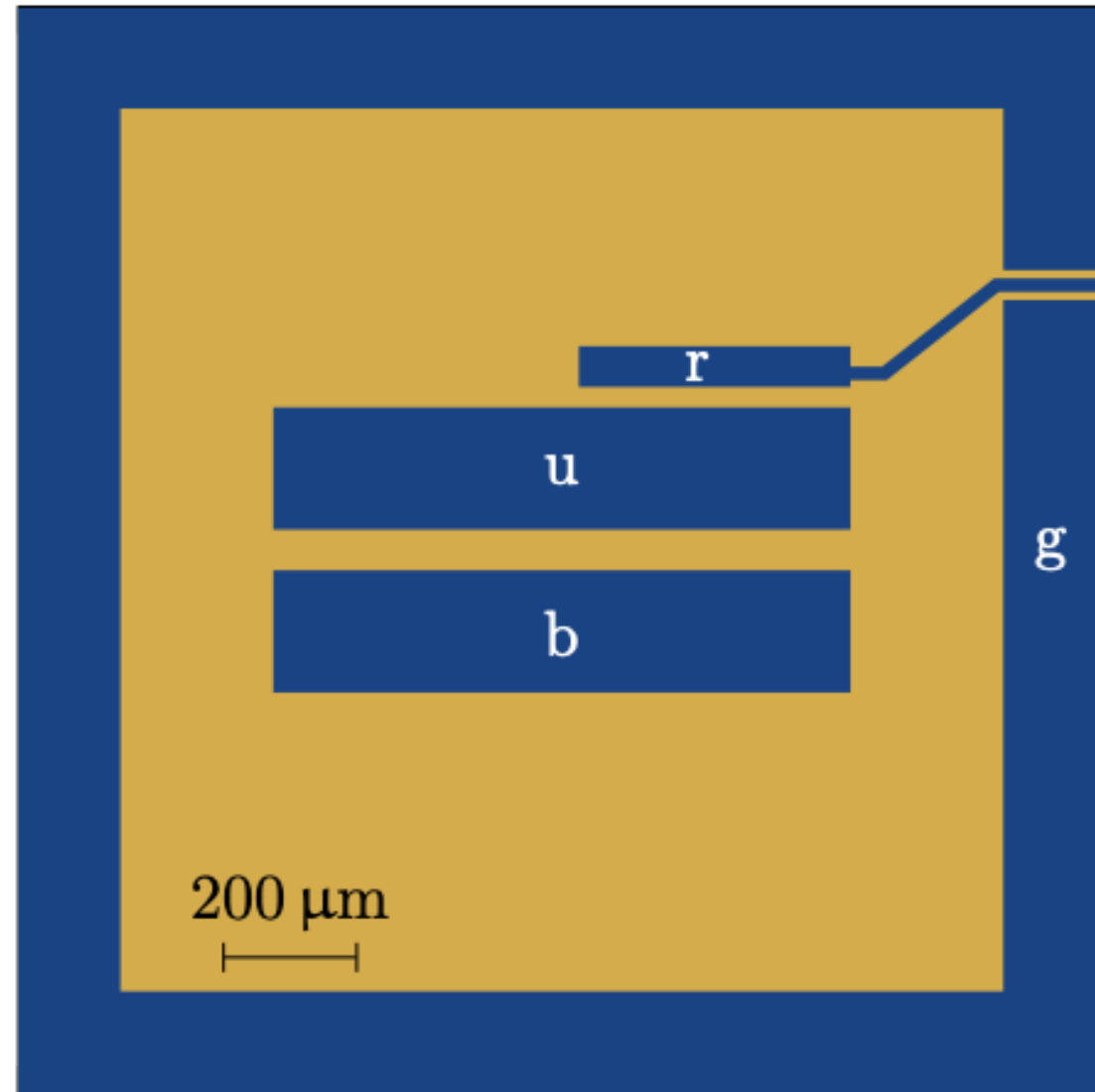
DoF: Degree of Freedom

Solver	DoF	Mode 1 (GHz)	Mode 2 (GHz)
Single Resonator			
Palace MMR	111,374,732	9.996	-
Palace AMR	69,301,924	9.993	-
Comsol	102,109,058	9.991	-
Ansys	51,411,402	10.004	-
Resonator Coupled to Feedline			
Palace MMR	74,348,288	7.095	-
Palace AMR	96,313,892	7.094	-
Comsol	122,062,996	7.092	-
Ansys	50,743,641	7.101	-
Transmon Coupled to Resonator			
Palace MMR	65,688,988	6.217	9.503
Palace AMR	95,143,250	6.225	9.507
Comsol	96,184,134	6.218	9.500
Ansys	42,708,384	6.221	9.507
Two Coupled Transmons			
Palace MMR	39,927,764	5.747	6.902
Palace AMR	62,522,530	5.754	6.911
Comsol	99,657,414	5.745	6.900
Ansys	41,449,212	5.754	6.911



CAPACITANCE EXTRACTION

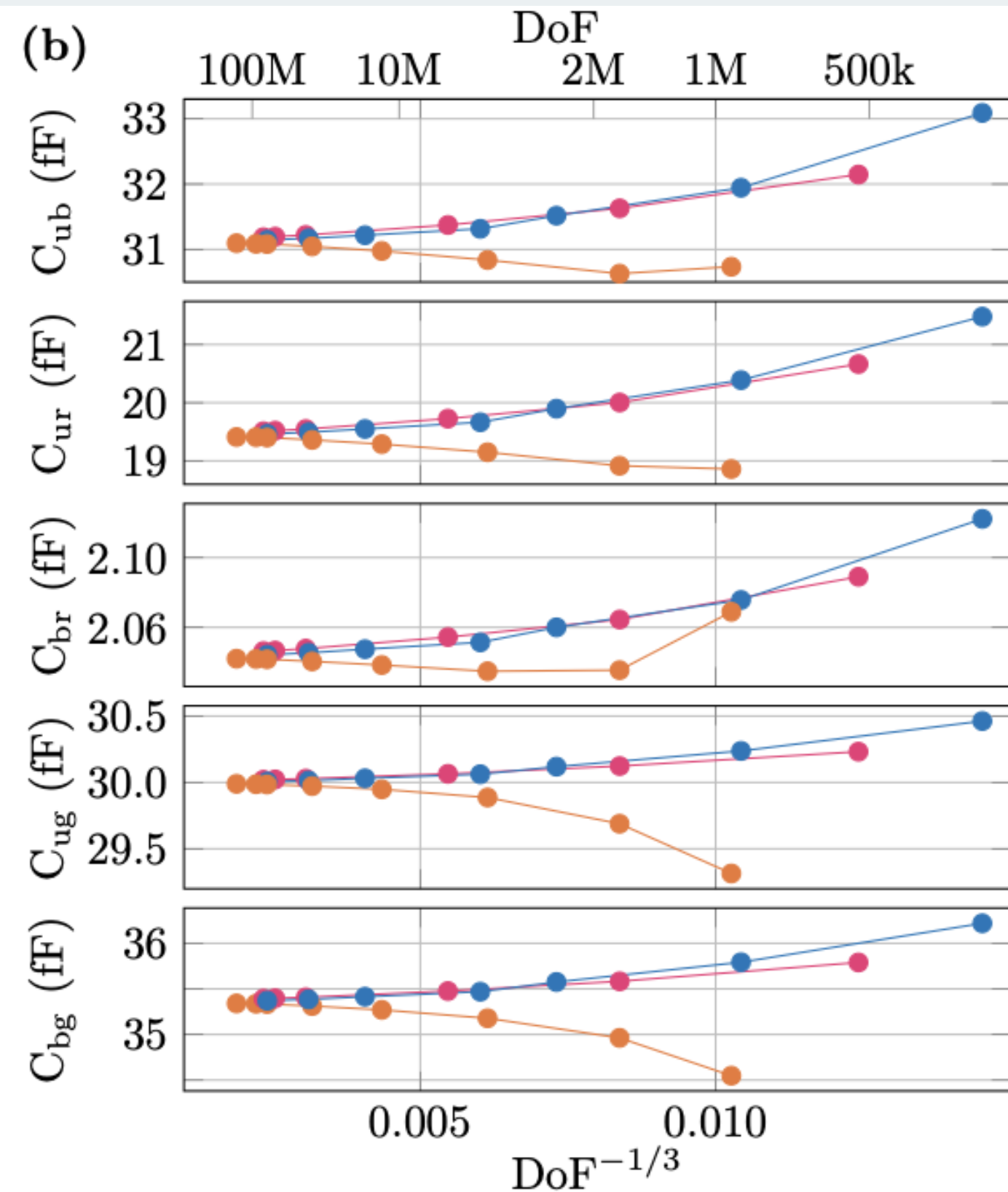
(a)



■ Metals (PEC)
■ Dielectric (Si)

—●— ANSYS (O2/AMR)
—●— COMSOL (O2)
—●— Palace (O2/AMR)

(b)



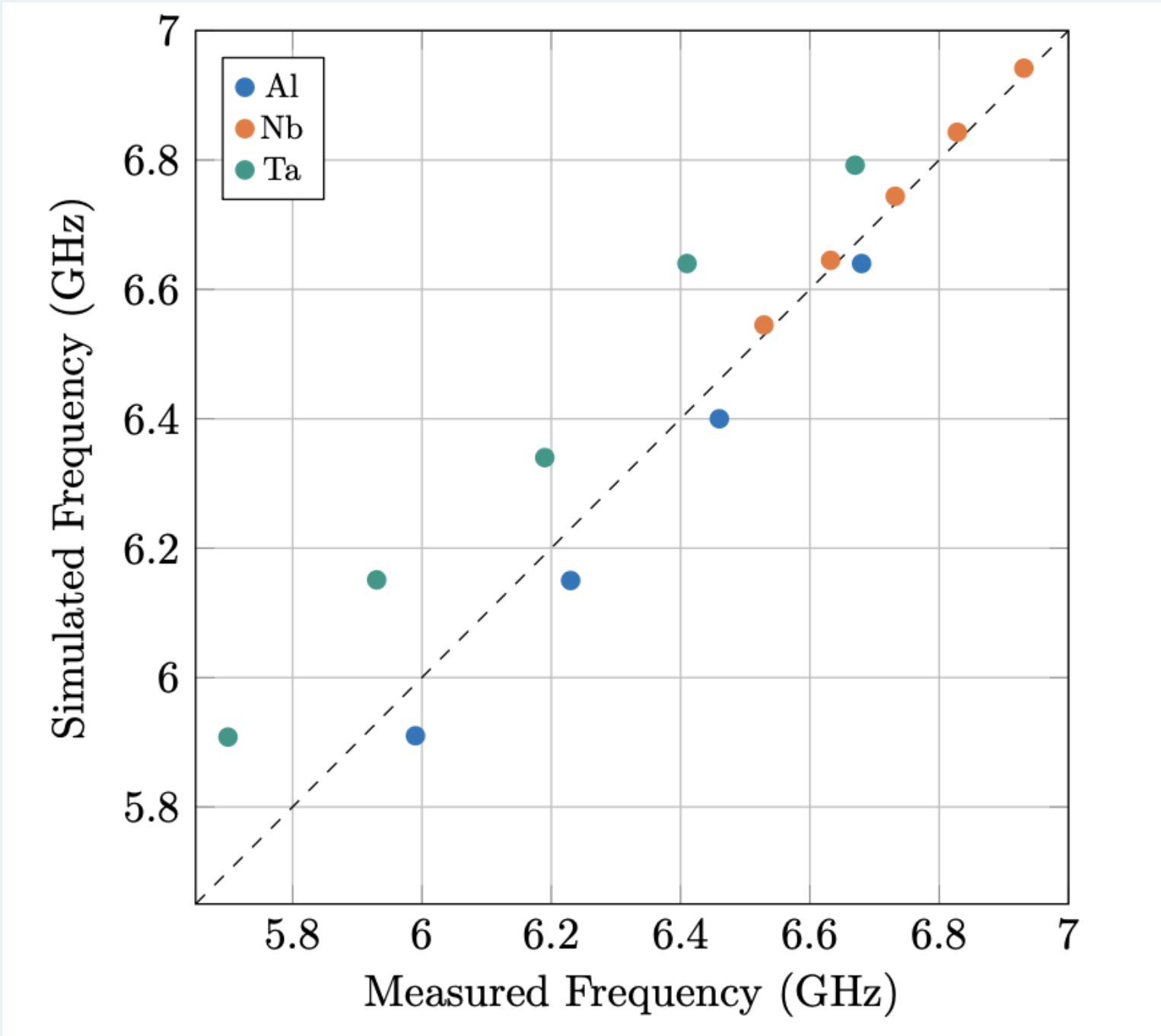
CAPACITANCE EXTRACTION

Table II. **Final capacitance simulation results.** Comparison of the final converged capacitance values and corresponding degrees of freedom (DoF) for each solver (Palace, COMSOL, and Ansys Q3D) from the electrostatic convergence study in Fig. 3.

Solver	DoF	C_{ub} (fF)	C_{ur} (fF)	C_{br} (fF)	C_{ug} (fF)	C_{bg} (fF)
Palace AMR O2	71,442,078	31.145	19.463	2.044	30.006	35.370
Comsol MMR O2	77,464,079	31.187	19.509	2.046	30.030	35.393
Ansys AMR O2	148,051,912	31.095	19.410	2.042	29.990	35.341

COMPARISITION SIMULATIONS VS EXPERIMENTAL RESULTS

RESONADORES



Resonator	Simulated (GHz)	Measured (GHz)	Difference (%)
Aluminum Resonators			
1	5.91	5.99	1.34
2	6.15	6.23	1.28
3	6.40	6.46	0.93
4	6.64	6.68	0.60
Niobium Resonators			
1	6.545	6.529	0.24
2	6.645	6.632	0.20
3	6.744	6.732	0.19
4	6.843	6.828	0.22
5	6.942	6.931	0.16
Tantalum Resonators			
1	5.91	5.70	3.52
2	6.15	5.93	3.59
3	6.40	6.19	2.37
4	6.64	6.41	3.46
5	6.79	6.67	1.80

COMPARISON SIMULATIONS VS EXPERIMENTAL RESULTS

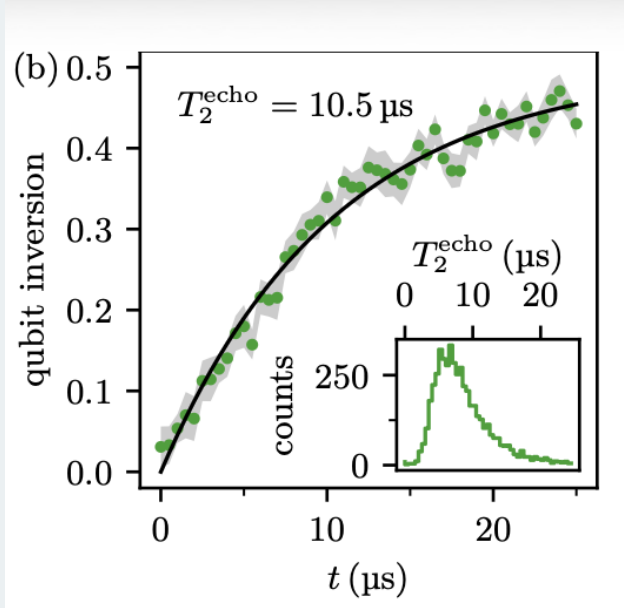
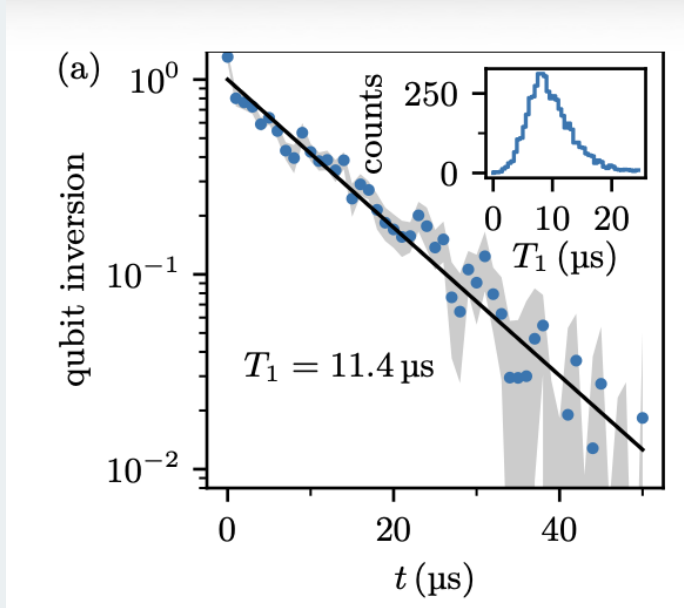
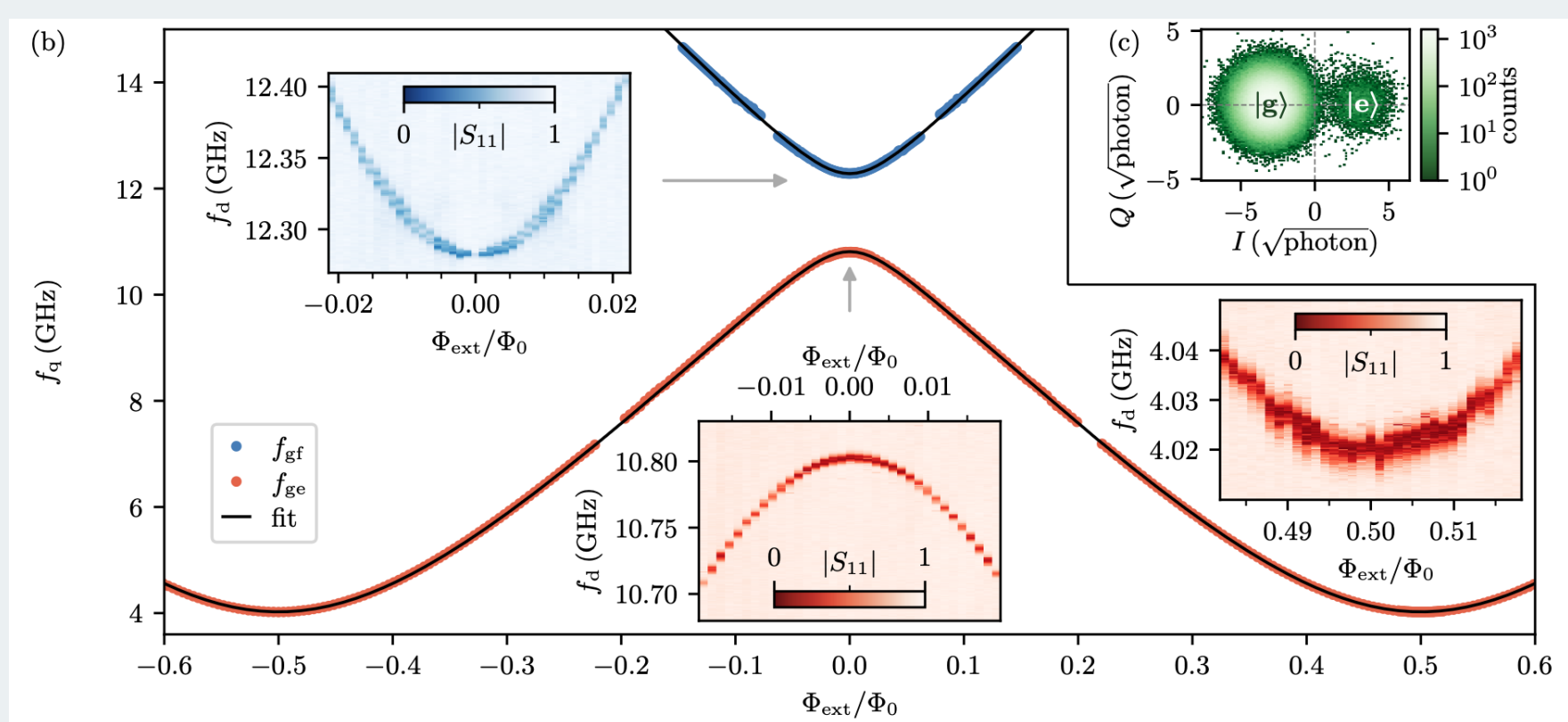
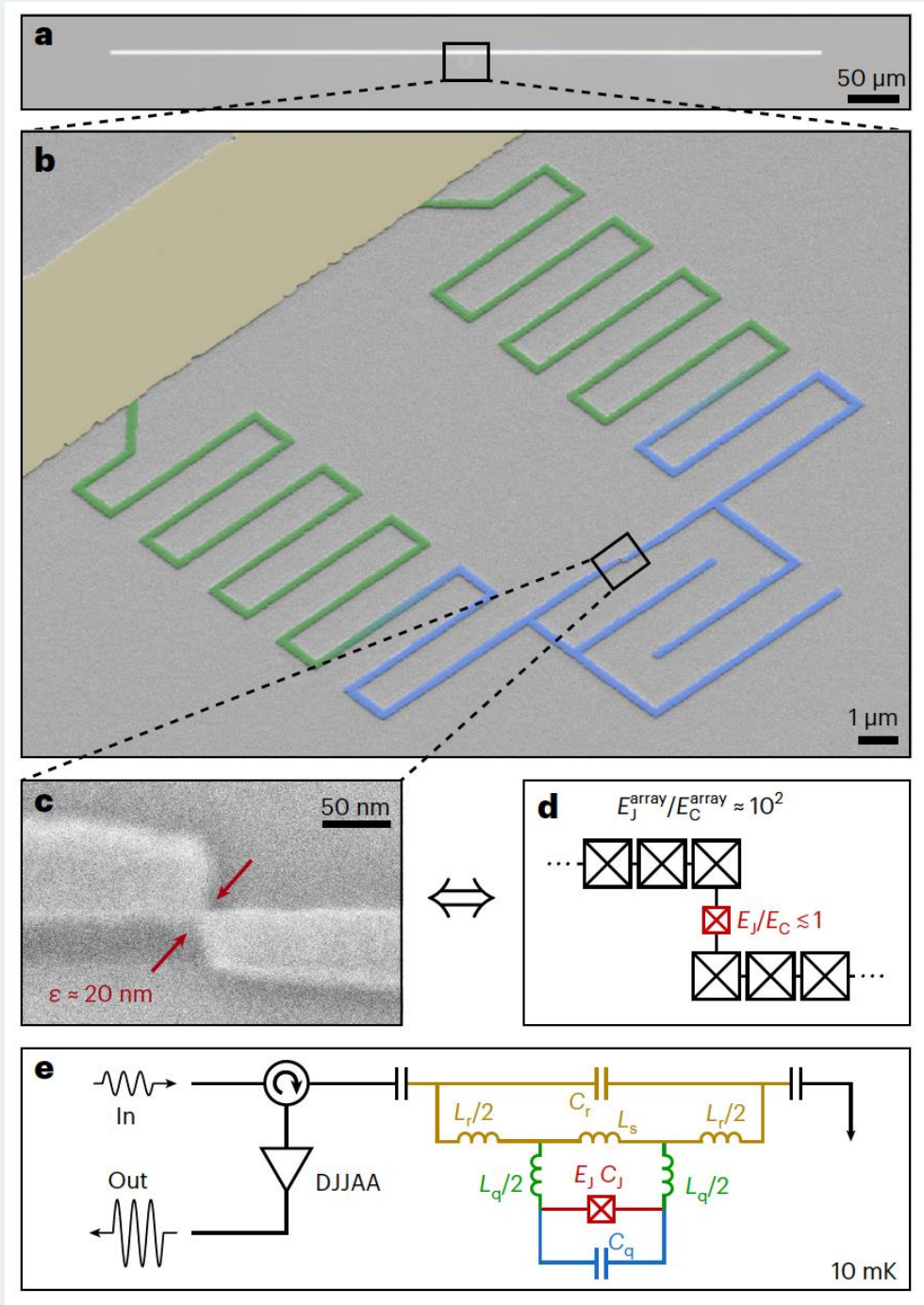
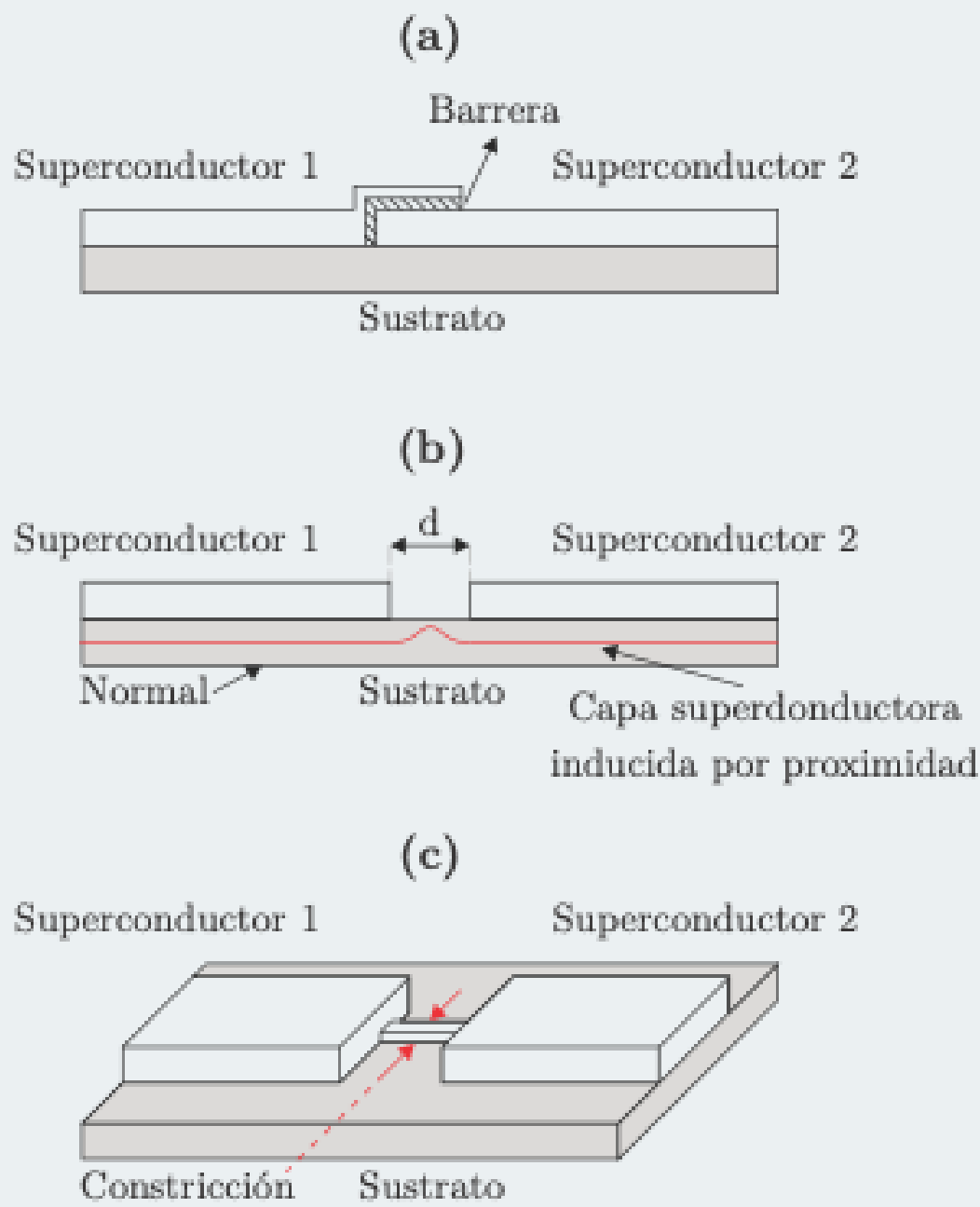
QUBIT

Table IV. **Comparison of simulated and experimental transmon parameters.** This table lists the qubit frequency (f_{ge}), qubit anharmonicity (α_q), resonator frequency (f_r), and coupling strength (g) obtained from simulation and experiment for each device. Simulated values are shown to the left of the measured values, with measured data given in square brackets. Qubits from Device A [27] and Device B [15] are labeled with the prefixes A and B, respectively.

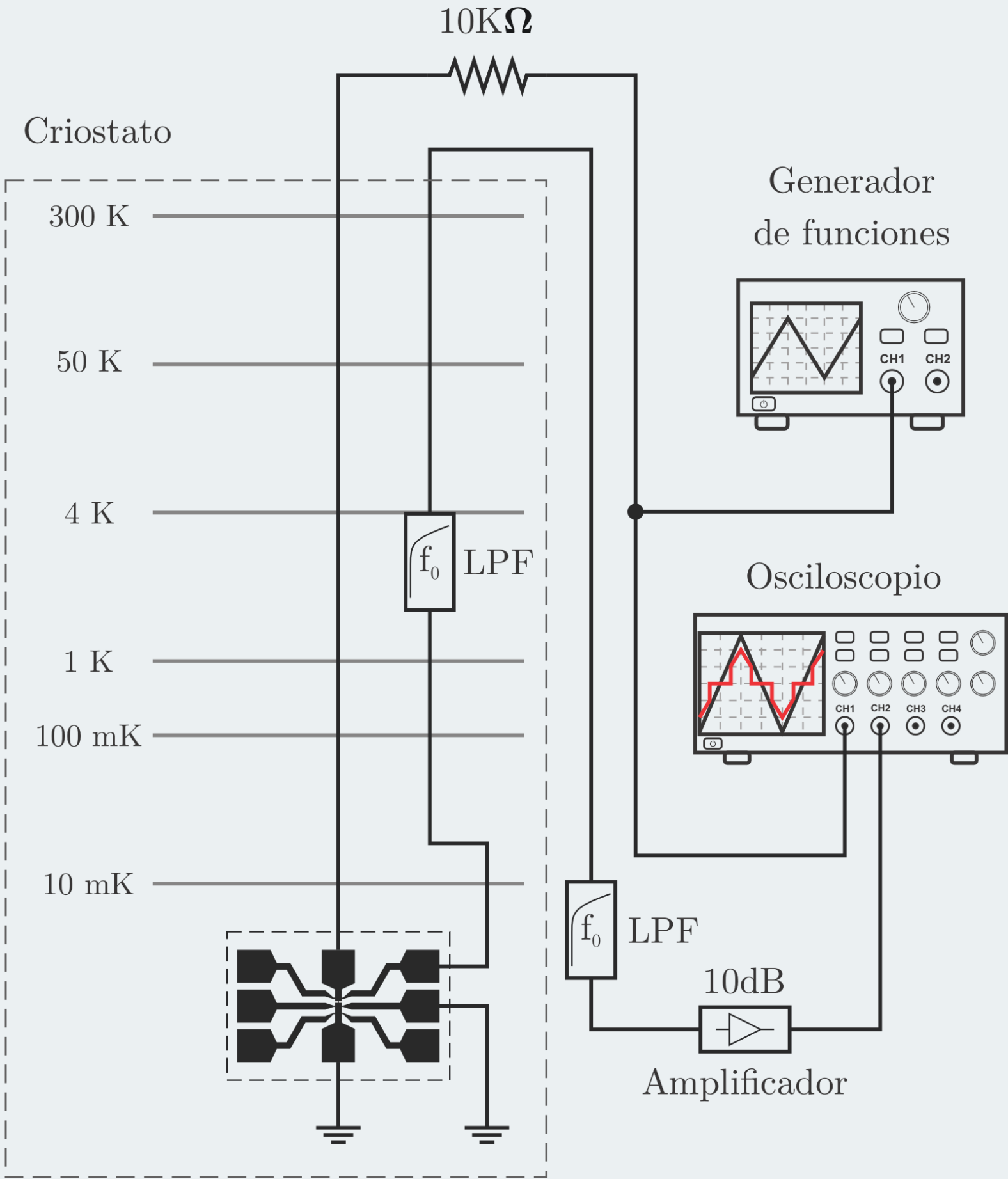
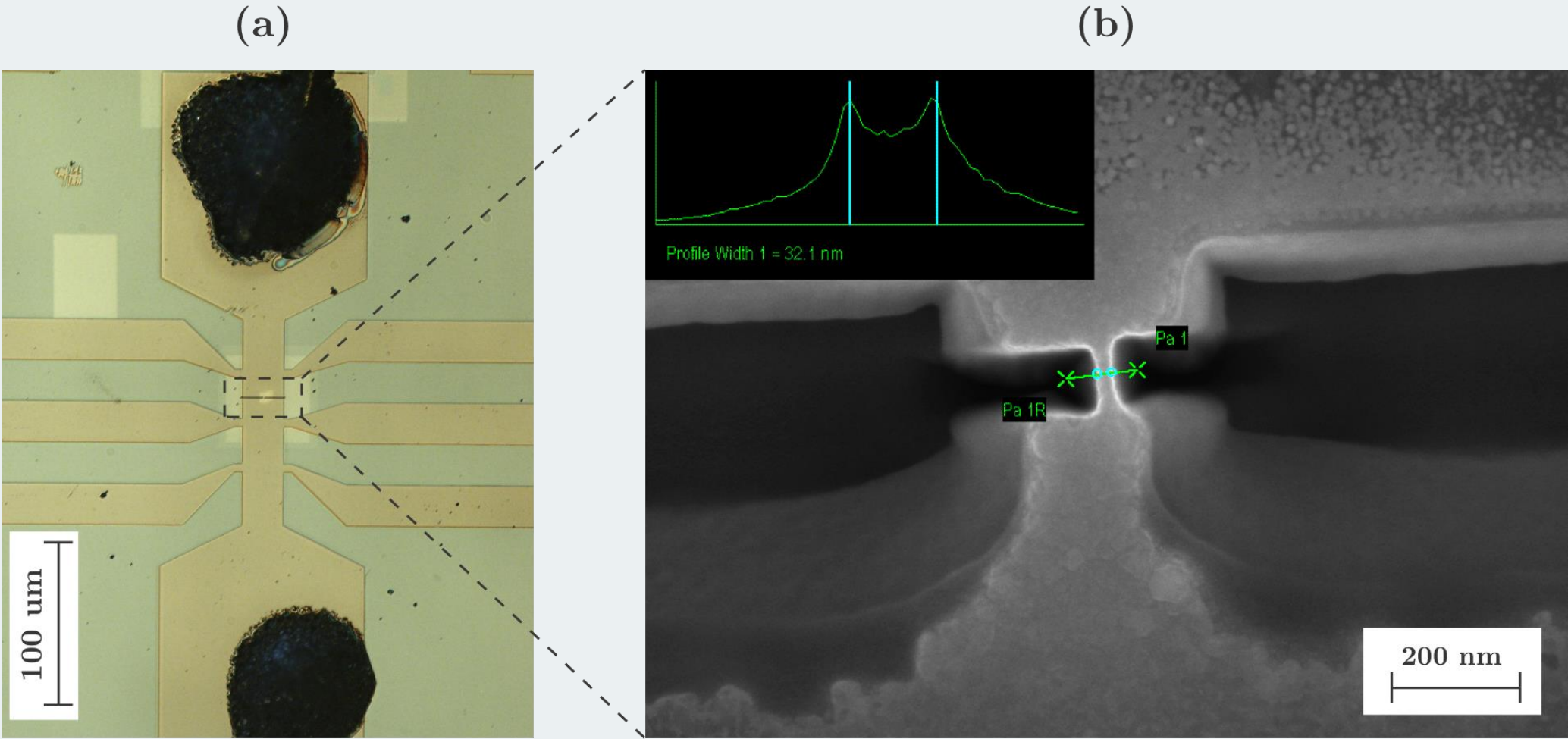
Device	f_{ge} (GHz)	$\alpha_q/2\pi$ (MHz)	f_r (GHz)	$g/2\pi$ (MHz)
AQ1	4.60 [4.20]	193 [212]	7.12 [6.94]	56 [60]
AQ2	5.13 [4.65]	171 [180]	7.27 [7.09]	55 [61]
AQ3	5.61 [5.37]	178 [140]	7.43 [7.21]	48 [54]
BQ1	4.18 [4.22]	144 [153]	6.34 [6.12]	75 [60]
BQ2	3.96 [3.90]	142 [154]	6.60 [6.35]	90 [66]
BQ3	4.33 [4.45]	173 [189]	6.72 [6.47]	88 [70]
BQ4	3.42 [3.59]	143 [164]	6.82 [6.57]	89 [66]
BQ5 ^a	3.85 [4.10]	178 [210]	6.96 [6.66]	66 [52]

MUCHAS GRACIAS

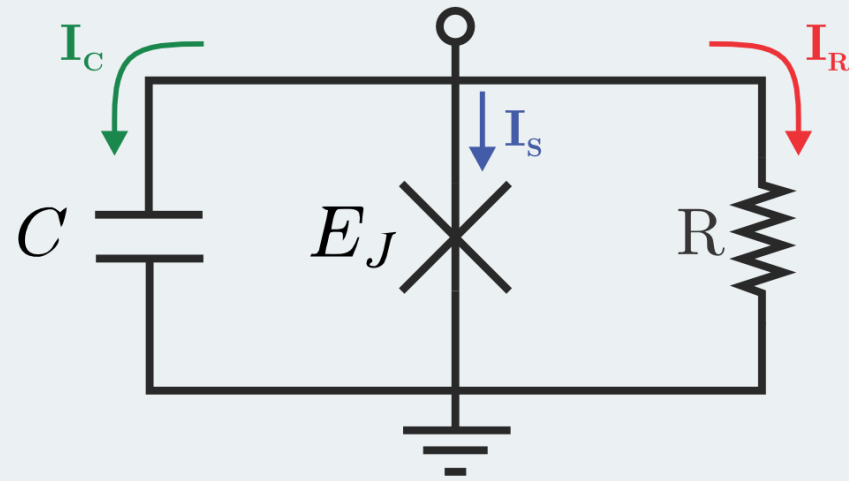
GRALMONIUM



CONSTRICCIÓN



MODELO RCJS

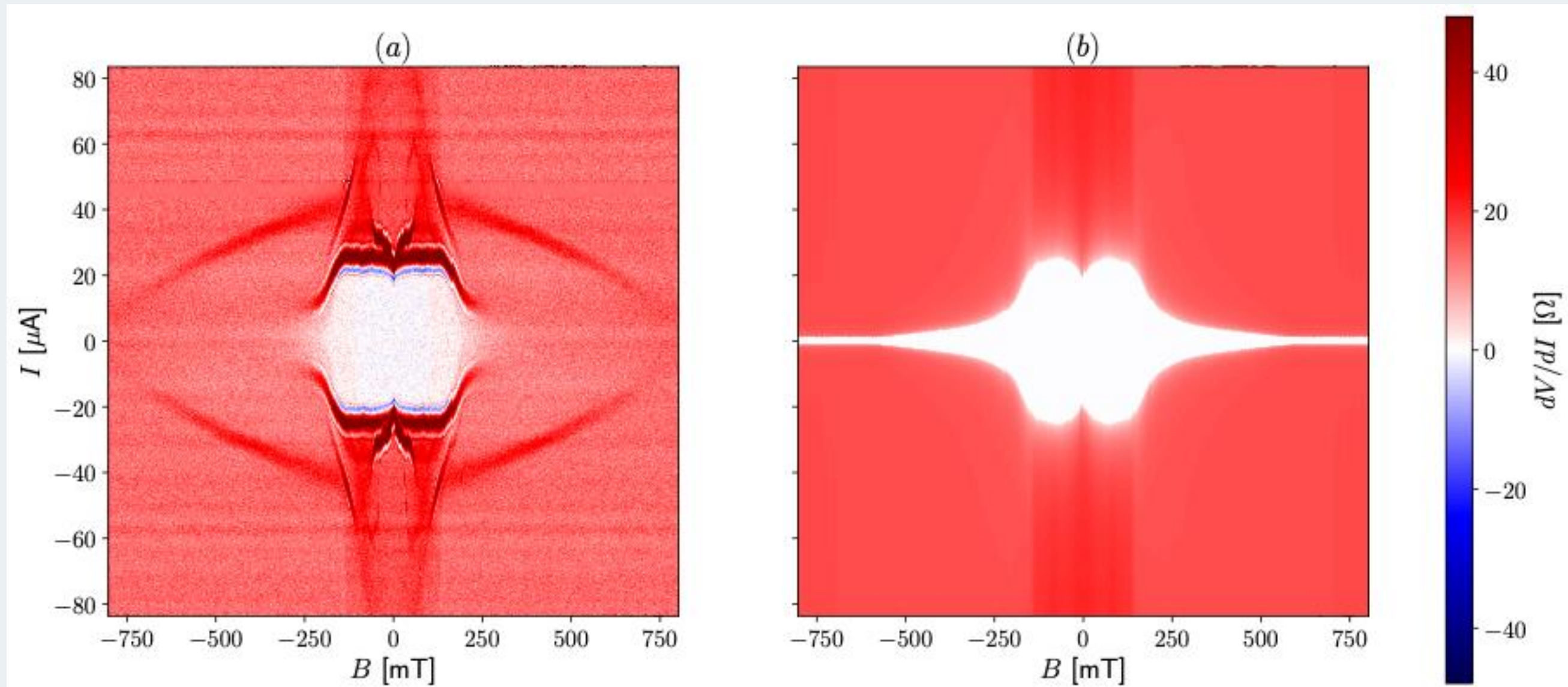


$$I = I_0 \sin(\delta) + \frac{\varphi_0}{R} \dot{\delta} + \varphi_0 C \ddot{\delta}$$

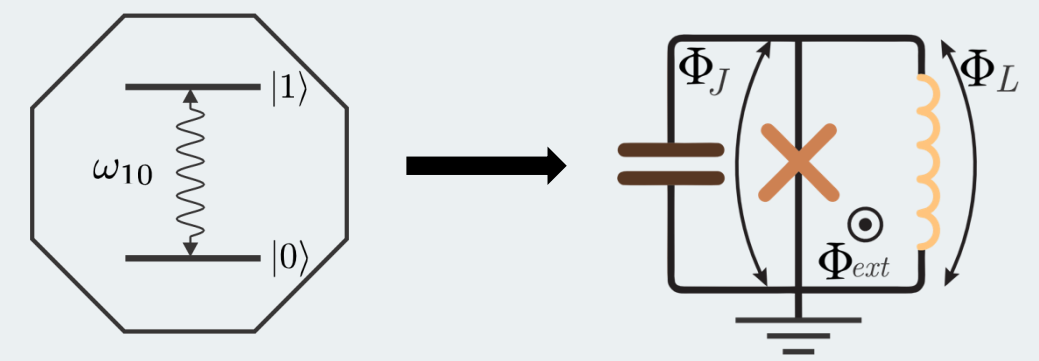
SOBREAMORTIGUADO

$$C \ll 1$$

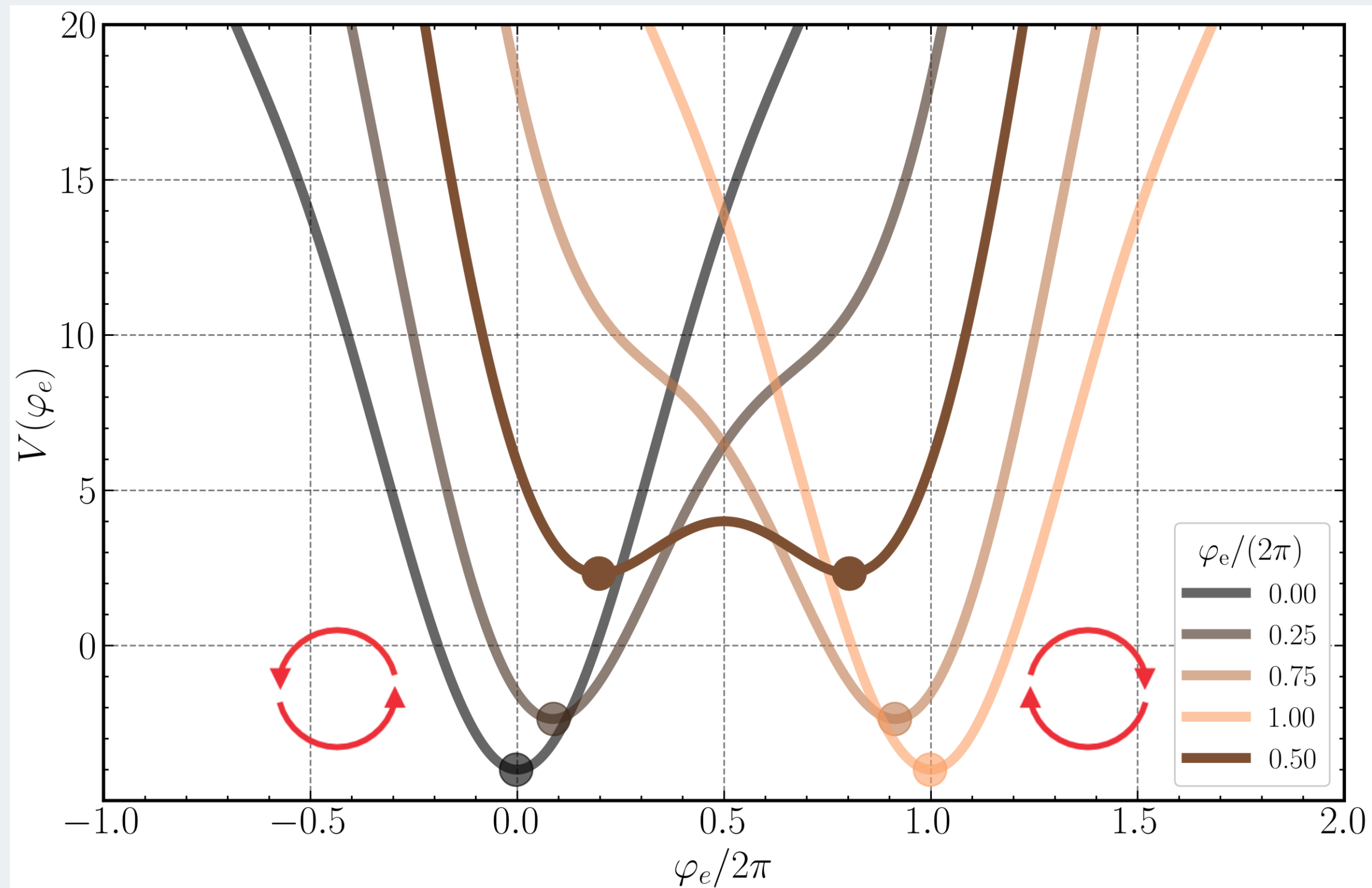
$$\dot{\delta} = \frac{I_0 R}{\varphi_0} \left(\frac{I}{I_0} - \sin \delta \right)$$



QUBIT DE FLUXONIUM



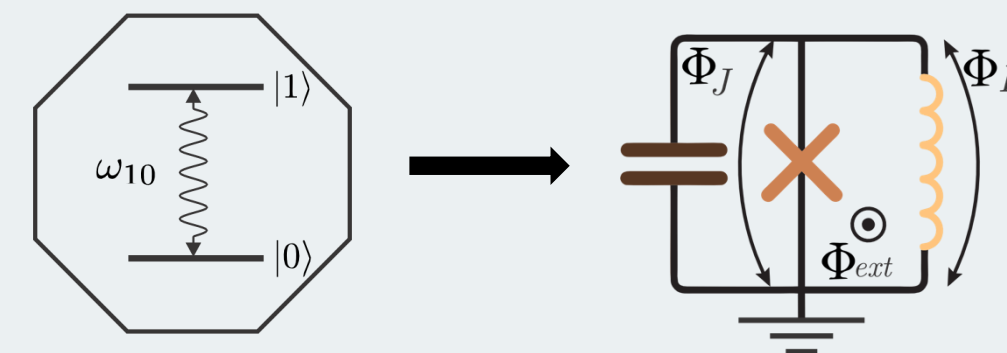
28



$$\mathcal{H} = 4E_C \hat{n}^2 - \underbrace{E_J \cos(\hat{\varphi}) + E_L (\hat{\varphi} - \varphi_e)^2}_V$$

$$\{E_J, E_C, E_L\} = \{4, 1, 1\} \text{ GHz}$$

QUBIT DE FLUXONIUM

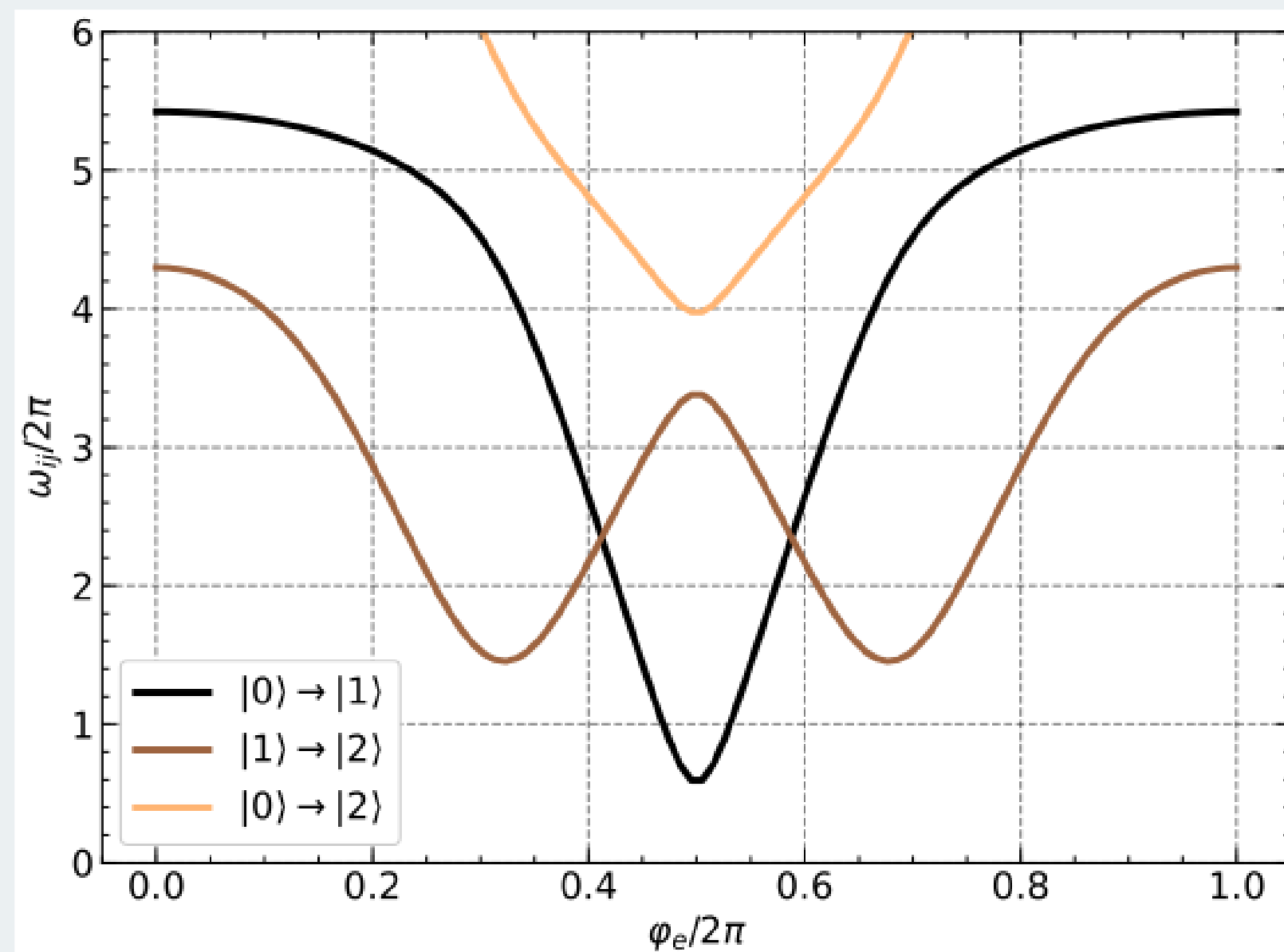


29

ANARMONICIDAD

$$\omega_{01}/2\pi = 0.58 \text{ GHz}$$

$$\omega_{12}/2\pi \text{ y } \omega_{02}/2\pi > 3 \text{ GHz}$$



REGLAS DE SELECCIÓN

$\hat{\varphi} \longrightarrow$ Base computacional.

$\hat{n} \longrightarrow$ Niveles excitados.

